Curvature Polarization Transport Gravity

A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions

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Abstract

Curvature Polarization Transport Gravity (CPTG) is a scale-structured geometric framework in which ordinary physical stress–energy remains the material source while curvature polarization and curvature transport supply additional geometric response. The Einstein term remains present throughout, and the gravitational theory returns continuously to General Relativity when the polarization–transport response becomes negligible. Galaxies, pressure-supported clusters, anisotropic dynamical systems, and compact objects are treated as symmetry-, state-, and scale-dependent realizations of this common source–response architecture rather than as separate gravity theories.

The resolved bound-system law combines the measured baryonic source with object-local support structure and a nonlinear polarization–transport response. Its analytic weak-field reduction contains the fourth-power baryonic Tully–Fisher scaling and the fixed-outer-source far-field law. Structural Mode N is a post-solution fingerprint of gravitational organization: its parent meaning is common across regimes, while the organization operator follows the geometry of the solved object. In the compact sector, CPTG admits a finite physical interior, separates localized source mass from the geometric contribution to the conserved exterior gravitational charge, retains the ordinary Kerr or Schwarzschild exterior, and evaluates rotating structural quantities only after the compact state has closed.

Cosmology contains one native π-branch. Acoustic, affine luminosity, CMB, BAO, growth, and lensing quantities are downstream projections or comparison coordinates of that native geometry rather than additional cosmological branches. The primordial nuclear sector realizes the same polarization–transport organization in a conserved reaction-state space: free nucleons, deuterium, mass three, and helium four form the vertex–bridge–closure–saturation backbone, with an exact rank-four reaction-source geometry propagated by native network dynamics. This article states the parent laws, authority hierarchy, overlap requirements, and derivational boundaries; numerical validation and regime-specific evidence are retained in the cited companion papers.

1 Introduction

CPTG is built around a simple physical distinction: ordinary matter, radiation, pressure, internal energy, neutrinos, electromagnetic fields, and other physically present sources remain on the material side of the theory, while the additional gravitational response is geometric. Curvature polarization describes how the sourced geometry reorganizes, and curvature transport describes how that organized response is retained, redistributed, or directionally carried through an extended or evolving system. CPTG therefore does not require a separate cold-dark-matter gravitational sector to supply the additional response attributed here to geometry.

The parent gravitational equation retains the Einstein tensor and adds polarization and transport on the geometric side. General Relativity is not discarded; it is the continuous limit reached whenever the additional CPTG response becomes negligible. The purpose of the reduced laws is to express this parent organization in the variables appropriate to the physical problem without changing the source–response ordering.

The gravitational realizations consequently follow the geometry and state of the object. Resolved disk galaxies use the measured baryonic source and local support state to close the nonlinear radial field. Pressure-supported clusters use the corresponding spherical support interface. Strongly anisotropic systems retain directional transport explicitly. Compact objects use a finite strong-field realization that closes a spherical mass state, admits a conditional mass–spin handoff, and joins the ordinary Kerr or Schwarzschild exterior in the common regime. These realizations share the parent architecture while retaining the coordinates required by their own symmetry and boundary data [9, 11].

Structural Mode N is downstream of those gravitational solutions. Its parent definition compares the structural extent of a solved object with a geometry-appropriate organization length, making N an object-specific structural fingerprint rather than a fit parameter or a universal acceleration scale. Disk galaxies, spherical clusters, and finite rotating compact states therefore use different organization operators while preserving the same post-solution meaning. CSMI is a downstream cataloging representation of the continuous galaxy fingerprint, not an input to the gravitational calculation [8].

Cosmology is organized by one native π-branch. The acoustic branch, affine luminosity map, CMB coordinates, standard-ruler coordinates, growth variables, and lensing variables are projections or comparison coordinates of that native state. They do not define additional CPTG cosmologies. The common-phase rule fixes how the branches remain synchronized, while each observational interface retains its declared role [12, 13, 15].

The nuclear realization is not obtained by applying the gravitational field equation to nuclei. It carries the polarization–transport organization into a conserved reaction-state geometry. A fixed scalar curvature response and exact reaction stoichiometry define the source drive; the native nuclear network then performs the dynamical pushforward. The light-sector backbone is organized as vertex, bridge, closure, and saturation, with a separate live mass-seven transport law and a declared boundary between native primordial dynamics and reduced heavy-sector continuation [16, 17, 19].

The common theoretical ordering is

$$
\begin{aligned}
&\rightarrow \quad \text{physical source and native geometric state} \\
&\downarrow \quad \text{geometric invariants, symmetry, and boundary/state data} \\
&\downarrow \quad \text{declared gravitational reduction or native state-space realization} \\
&\downarrow \quad \text{polarization–transport dynamics or native dynamical pushforward} \\
&\downarrow \quad \text{solved physical state} \\
&\downarrow \quad \text{observational or network projection} \\
&\rightarrow \quad \text{reported comparison quantity.}
\end{aligned}
\tag{1}
$$

The direction of this hierarchy is part of the theory. Downstream observations do not acquire native authority merely because they are useful for comparison, and quantities extracted after solution do not feed backward into the state unless the defining reduction explicitly requires them. Cross-regime consistency is enforced through conservation, declared authority, and agreement in genuine overlap limits. The detailed numerical tests that establish particular realizations are contained in the companion publications; this parent article states the laws those tests address.

The sections that follow give the variational foundation and authority hierarchy, the resolved and asymptotic gravitational laws, Structural Mode, pressure-supported and anisotropic dynamics, finite compact gravity, the native cosmological branch and its projection architecture, the primordial nuclear reaction-state law, and the cross-regime consistency conditions that bind the framework together.

2 Variational Foundations and Theory Hierarchy

CPTG organizes gravitational response through a coupled hierarchy of General Relativity, curvature polarization, and curvature transport. These components are computed differently but operate together as one geometric gravitational framework. Gravitational sectors are registered reductions or realizations of one CPTG gravity-side architecture within their declared symmetry, scale, state, and boundary domains; non-gravitational sectors may realize the same curvature–polarization–transport organization in their own native state spaces.

2.1 Scale-Structured Realization Principle

CPTG is a framework of physical scales. The common theory fixes the source–response ordering, the polarization–transport architecture, the conservation requirements, and the rule that a physical realization is selected before its downstream observable is constructed. The operative reduced law is the law appropriate to the source, physical scale, symmetry, dynamical state, and boundary or state data of that problem.

A galaxy, coherent cluster, anisotropic merger, compact structure, cosmological background, and conserved nuclear reaction state therefore need not share one reduced component equation. Their consistency is determined by derivation from the common CPTG hierarchy and by agreement wherever two reductions possess a genuine common physical limit. If two realizations A and B overlap on a physical domain $\mathcal{O}$, every quantity Q that has the same physical meaning in both descriptions must satisfy

$$Q_A|_{\mathcal{O}} = Q_B|_{\mathcal{O}}\,. \tag{2}$$

Equation (2) is the cross-scale matching rule. It requires shared physical content to agree in the regime where both reductions apply. The GR recovery limit, continuity of structural quantities within a bound-system reduction, and inheritance of one cosmological native state by its downstream projections are instances of this same ordering.

2.2 Curvature Polarization and Curvature Transport

The gravitational sector of CPTG is formulated as a nonlinear geometric response theory defined by a single action-based framework. At the parent gravity level the action is

$$S_{\mathrm{CPTG}} = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + \mathcal{L}_{\mathrm{pol}} + \mathcal{L}_{\mathrm{trans}} + \mathcal{L}_{\mathrm{m}}\right]. \tag{3}$$

Here $\mathcal{L}_{\mathrm{m}}$ contains the ordinary physical stress–energy source. The additional terms are geometric: $\mathcal{L}_{\mathrm{pol}}$ organizes the nonlinear polarization response of curvature to the sourced field, while $\mathcal{L}_{\mathrm{trans}}$ governs directional transport or redistribution of organized curvature structure. Equation (3) states the common variational architecture of the gravitational sector. The explicit sector formulas used below are reduced representatives in the variables appropriate to each physical realization, rather than one scalar expression imposed on every gravitational state. The nuclear reaction-state construction shares the polarization–transport organization but is not governed by Eq. (3) or its gravitational field equation.

For later conservation statements it is useful to define the geometric response tensors directly by functional differentiation. With

$$S_{\mathrm{pol}} = \int d^4x\sqrt{-g}\,\mathcal{L}_{\mathrm{pol}}, \qquad S_{\mathrm{trans}} = \int d^4x\sqrt{-g}\,\mathcal{L}_{\mathrm{trans}}, \tag{4}$$

and the matter convention

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\mathrm{m}}}{\delta g^{\mu\nu}}, \tag{5}$$

the CPTG response tensors are defined by

$$\Pi^{\mathrm{pol}}_{\mu\nu} = \frac{16\pi G}{c^4}\frac{1}{\sqrt{-g}}\frac{\delta S_{\mathrm{pol}}}{\delta g^{\mu\nu}}, \tag{6}$$

$$\Theta^{\mathrm{trans}}_{\mu\nu} = \frac{16\pi G}{c^4}\frac{1}{\sqrt{-g}}\frac{\delta S_{\mathrm{trans}}}{\delta g^{\mu\nu}}. \tag{7}$$

These definitions are exact at the parent variational level even where the component forms of the two tensors are not explicitly reduced for a particular geometry.

Writing the field equation in terms of these variational responses gives

$$G_{\mu\nu} + \Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{8}$$

The right-hand side remains the ordinary physical stress–energy source. The two additional tensors on the left are geometric response terms. They are not dark-sector stress–energy tensors and are not independently assigned matter fluids. Their role is to encode how the curvature response depends on more than the pointwise amplitude of the source field.

This separation fixes an important interpretation of the theory: enhanced gravitational response and additional material source are not synonymous. If a solved CPTG field exceeds the direct baryonic response, that excess is not entered by adding a new contribution to $T_{\mu\nu}$; within the CPTG description it is carried by the geometric polarization–transport response on the left-hand side of Eq. (8). Thus "more gravitational response" does not by itself imply "more gravitating matter".

The relationship to General Relativity (GR) is explicit. Wherever the polarization–transport response becomes negligible,

$$\Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} \to 0, \tag{9}$$

Eq. (8) reduces continuously to the Einstein equation with the same material source. This recovery limit is one end of a continuous response hierarchy rather than a boundary between two unrelated theories. Schematically,

$$\begin{aligned} \Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} \simeq 0 \quad &\Longrightarrow \quad \text{GR-dominated regime,} \\ \Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} \text{ non-negligible} \quad &\Longrightarrow \quad \text{polarization–transport-significant regime.} \end{aligned} \tag{10}$$

As the polarization–transport response grows, it can become the dominant contribution governing the departure from the GR-only solution in the regime where that organized response becomes significant. This does not remove or replace the Einstein curvature term. In black-hole physics specifically, GR remains the controlling description for the exterior spacetime, the trapping horizon, orbital dynamics, and the ordinary relativistic evolution across most of the solution. The polarization–transport components become dynamically significant only in the extreme compact regime where the GR-only continuation would otherwise approach an infinite-curvature endpoint that is not physically admissible in CPTG. The reverse transition is equally continuous: where polarization and transport weaken, the solution moves smoothly back toward the GR-dominated limit of Eq. (9). There is therefore no discontinuous transition from one gravity theory to another.

The physical interpretation is correspondingly conservative: CPTG preserves the Einsteinian identification of gravity with geometry. General Relativity is the Einstein component of the coupled gravitational response, while curvature polarization and curvature transport provide additional geometric organization where their contributions become significant. These components operate together within one gravitational framework rather

than as replacements for one another. GR dominates wherever the polarization–transport contribution is negligible. For a black hole, the event or trapping horizon is not itself a boundary at which polarization–transport terms become significant: the horizon and most exterior strong-field physics remain GR-controlled, while the polarization–transport terms become important only in the deeper regime where a GR-only continuation would otherwise run to an inadmissible infinite-curvature endpoint. When the polarization–transport contribution again becomes negligible, the ordinary Einstein equation is recovered.

Equation (9) is the exact formal recovery statement. Precision recovery in a system where GR is already tightly tested is a separate empirical requirement: after the standard source and ephemeris parameters are treated consistently, any residual CPTG polarization or transport signal must remain below the relevant observational limits [6, 7]. This distinction prevents the existence of the GR limit from being mistaken for a claim that every precision-limit test has already been completed.

The contracted Bianchi identity constrains the same unified field equation. If the ordinary material source is independently conserved,

$$\nabla^{\mu} T_{\mu\nu} = 0, \tag{11}$$

then Eq. (8) requires

$$\nabla^{\mu} \left(\Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} \right) = 0. \tag{12}$$

This is a parent-level consistency condition on the combined polarization–transport response generated by the action. It requires conservation of the sum of the two geometric responses but does not require the polarization and transport tensors to be separately conserved.

The Noether requirement also fixes the status of any auxiliary geometric fields used to construct the response. Let Ψ_A denote any independent direction, state, or order field appearing in $S_{\mathrm{pol}} + S_{\mathrm{trans}}$, with Euler–Lagrange residual

$$\mathcal{E}_A \equiv \frac{1}{\sqrt{-g}} \frac{\delta (S_{\mathrm{pol}} + S_{\mathrm{trans}})}{\delta \Psi_A}. \tag{13}$$

For a diffeomorphism-invariant parent response action, the associated Noether identity reduces to Eq. (12) when every independent auxiliary field is on shell, $\mathcal{E}_A = 0$, or when the apparent auxiliary object is instead constructed covariantly from the solved geometry and therefore is not an independent background field. Consequently, a transport direction may be held fixed only after a declared reduction. At parent level a prescribed external S^{α} would not by itself provide the claimed automatic Noether closure; the direction must be geometry-selected or dynamically constrained. The same rule applies to any spacetime-varying suppression or state field. Structure-specific reduced quantities such as $a_{\star}$ and κ_{red} may enter as derived solution scales or boundary/reduction data, but they are not universal parent couplings that can be assigned arbitrarily as coordinate-dependent backgrounds.

Equations (7)–(13) close the parent conservation logic. Their arbitrary-geometry component realization depends on the chosen covariant geometric fields; the expressions below

are the reduced representatives used in the declared physical sectors.

A limited covariant bridge is available in a static region where a hypersurface-orthogonal Killing field remains timelike. For the normalized static congruence, the proper acceleration is the spatial gradient of the logarithm of the Killing lapse. Here $\mathfrak{a}_\mu$ denotes the geometric four-acceleration in inverse-length units, so that $c^2\mathfrak{a}_i \to g_i$ in the weak-field limit; its projected gradient correspondingly reduces to the spatial gradient of the Newtonian field. Consequently a static covariant scalar can reproduce the reduced field-plus-gradient structure

$$\mathcal{C}_{\rm stat} = c^4\left[\mathfrak{a}_\mu\mathfrak{a}^\mu + \kappa_{\rm stat}^2(D_\mu\mathfrak{a}_\nu)(D^\mu\mathfrak{a}^\nu)\right] \longrightarrow g_j g_j + \kappa_{\rm stat}^2(\partial_i g_j)(\partial_i g_j) \tag{14}$$

in the declared static weak-field regime. This bridge is included only to show that the reduced field-plus-gradient structure is not intrinsically tied to Cartesian notation. It is not used as a black-hole interior invariant and is not extended through a Killing horizon, where the normalized static congruence is no longer a regular global description. For a static high-gradient scalar reduction with $a_\star > 0$, define

$$C_{\rm red} = g_j g_j + \kappa_{\rm red}^2(\partial_i g_j)(\partial_i g_j), \qquad \chi_{\rm red} = \frac{C_{\rm red}}{a_\star^2}. \tag{15}$$

Here $\kappa_{\rm red}$ is the gradient length selected by the declared reduction; it is not a universal identification of the galaxy-fixed $\kappa_{\rm struct}$ with the compact-sector $\kappa_{\rm c}$. The corresponding polarization representative is

$$\mathcal{L}_{\rm pol}^{(\rm red)} = \beta a_\star^{2/3} C_{\rm red}^{1/3} = \beta a_\star^{4/3}\chi_{\rm red}^{1/3}. \tag{16}$$

For a declared organized transport direction S^α and a sector-appropriate suppression function $\Sigma_{\rm red}$, the explicit directional transport representative is

$$\mathcal{L}_{\rm trans}^{(\rm red)} = -\frac{\lambda_t}{2}\left(S^\alpha\Sigma_{\rm red}\nabla_\alpha\Phi\right)^2. \tag{17}$$

Equations (16) and (17) expose the shared reduced variational content: polarization is gradient-sensitive and transport is directional and suppression-weighted. The quantities $\kappa_{\rm red}$, S^α, $\Sigma_{\rm red}$, and the effective normalizations that appear after reduction take the forms registered for the physical state, symmetry, and scale of the problem.

Physical state, symmetry, scale, and boundary data determine the operative sector reduction and its variables. Downstream observational fits test that reduction; they do not retrospectively redefine the assumptions or authority of the upstream construction.

2.2.1 Coupling and Transport-Direction Status

The transport direction S^α is a normalized geometric direction field, $S^\alpha S_\alpha = 1$, in the reduced sectors where an explicit direction is retained. Consistent with Eq. (13), it may be held fixed during a declared reduced variation only after the physical reduction has selected it. At parent covariant level it must either be constructed from the solved geometry/state

or be promoted to an independent geometric field with its own equation of motion and normalization constraint. It is not an added matter four-vector and it is not assumed to be one universal preferred direction shared by every system. Different geometries may select different effective transport directions; a reduction must state the direction or the state information from which it is obtained.

The parent couplings also have a definite dimensional status. In the SI convention used here, with each Lagrangian density measured as an energy density, $[C_{\rm red}] = \mathrm{m}^2\,\mathrm{s}^{-4}$, $[\nabla_\alpha \Phi] = \mathrm{m}\,\mathrm{s}^{-2}$, and dimensionless S^α and $\Sigma_{\rm red}$, dimensional consistency requires

$$[\beta] = \mathrm{kg}\,\mathrm{m}^{-7/3}\,\mathrm{s}^{2/3}, \qquad [\lambda_t] = \mathrm{kg}\,\mathrm{m}^{-3}\,\mathrm{s}^2. \tag{18}$$

These are parent-action couplings, not extra matter densities. In reduced galaxy calculations their normalization is absorbed into the effective response, whereas directional merger or compact reductions may retain the corresponding coupling structure explicitly. Absorbing their normalization into a reduction-specific coefficient does not alter their parent-action status, and such a reduced coefficient is not identified with either parent coupling unless that identification is explicitly derived. The distinction reflects the sector reduction rather than a change of theory.

The distinction between polarization and transport is physical. Curvature polarization answers the local source–response question: how does curvature reorganize in response to the strength, gradient, and structure of the baryonic source? Curvature transport answers the subsequent dynamical question: how is that organized response carried, retained, redistributed, or directionally displaced through an extended or changing system? In a quasi-static system polarization can generate enhanced support while preserving recovery of the high-acceleration baryonic limit. When the geometry is approximately radial and slowly evolving, much of the transport sector can reduce to a structural remnant; when the geometry is strongly anisotropic or dynamical, as in a dissociative cluster merger, transport can remain explicit and shift the curvature-supported observable response relative to the collisional baryonic gas.

Observational projection is downstream of physical curvature transport. Once a native or reduced CPTG state is solved, a CMB, baryon acoustic oscillation (BAO), abundance, lensing, or other comparison map expresses that state in the required reporting coordinate without supplying an additional physical transport mechanism. Cosmological projections therefore remain responses of one native history, while the nuclear source geometry is propagated by the native reaction network without replacing its microscopic rates or dynamics.

2.3 Known Cross-Scale Realizations

CPTG realizations span resolved weak-field systems, anisotropic dynamical systems, finite compact states, native cosmology, and conserved nuclear reaction-state geometry. This breadth does not require one reduced component equation to be copied across scales. The common content is the source–response hierarchy, conservation, overlap consistency, and the distinction between curvature polarization, curvature transport, and downstream

readouts; the operative variables and reduced equations follow the physical scale, symmetry, support state, and boundary data of the problem.

Resolved galaxy	observed baryonic disk source plus rotational support → registered support-conditioned radial CPTG realization, with the structural acceleration evaluated locally across the resolved field and structural organization measured only after the solution is fixed.
Pressure-supported cluster	observed baryonic/thermal source plus hydrostatic or Jeans support → spherical support-conditioned realization of the resolved CPTG field architecture; independent reconstructions remain independent solved states and retain separate post-solution structural fingerprints.
Dissociative merger	anisotropic source geometry → polarization plus an explicit directional transport operator along the retained S^{α}, rather than forcing the dynamical state into a quasi-static spherical reduction.
Compact black hole	finite spherical mass state → closed source–transport–polarization mass continuation, conditional mass–spin handoff, and Kerr-compatible rotating post-solution readouts, with the ordinary GR/Kerr exterior retained in the overlap domain.
Cosmology	universe-scale curvature/transport state → one native π-branch, from which acoustic, luminosity, CMB, growth, lensing, and other observational comparison coordinates are projected at common geometric phase.
Nuclear reactions	shared CPTG response geometry plus exact conserved reaction-state directions → network-native dynamical pushforward while the microscopic reaction network retains its own rates and conservation structure.

These entries summarize developed realizations of one CPTG architecture without promoting any regime-specific coordinate representation to a universal formula. Resolved galaxies and pressure-supported clusters use support-conditioned weak-field reductions; anisotropic mergers retain directional transport explicitly; compact black holes use a relativistic finite mass-state continuation; cosmology begins from one native background before observational projection; and the nuclear realization is propagated in its own conserved reaction-state space rather than by the gravitational field equation.

For calculations, the physical regime must be specified before its nonlinear equations are solved. The operative direction remains the hierarchy in Eq. (1): source geometry and state data are fixed first, the symmetry/scale/state realization is declared second, the corresponding polarization–transport dynamics or native pushforward is solved third, and only then is the solved state translated into a reported observable, structural fingerprint, or network coordinate.

This ordering prevents a convenient scalar formula from being mistaken for the framework

as a whole. Compatibility is assessed within each declared realization by its mathematical consistency, physical domain, boundary/state assumptions, and agreement with the common CPTG architecture in every genuine overlap domain.

Once this distinction is made, quantities must be distinguished according to whether they belong to the native theory or arise only after reduction, projection, conditioning, or numerical validation.

2.4 Native Geometry, Reduced Laws, and Comparison Coordinates

CPTG uses a strict hierarchy of quantity classes because the same symbol can otherwise acquire different physical meanings when it is translated between theory and observational pipelines. The classification is summarized in Table 1.

Class	Meaning	Preferred claim language
Native invariant	Quantity belonging to the defining CPTG geometric structure before observational conversion	fixed / defines
Exact derived quantity	Algebraic or integral consequence of native quantities	derived / follows
Reduced physical law	Regime-specific equation obtained after declared symmetry, scale, or state reduction	reduces to / is represented by
Closed realization	Mathematical continuation or reduction completed within its declared geometry and boundary domain	closes / is solved within
Conditional closure	Relation fixed only after an explicitly stated handoff, boundary, or conservation condition is imposed	holds under / follows given
Post-solution fingerprint or readout	Quantity measured from a completed solved state and not used to select that state	measures / is read from
Projected coordinate	CPTG result mapped into the coordinate used by an observational or numerical sector	maps to / is projected to
Data-conditioned coordinate	Quantity whose normalization or endpoint explicitly inherits an external observational scaffold	conditioned on / constructed from
Frozen comparison input	Upstream numerical setting held fixed before a declared comparison	held fixed / supplied as input
Validated result	Numerical or observational test outcome inside a stated domain and gate	passes / is reproduced
Bounded or diagnostic quantity	Relation or comparison object not fixed by the parent derivation	indicates / is not determined by

Table 1: Parent-level CPTG quantity classes. The class fixes the strongest permissible claim language and prevents a reduced, conditional, post-solution, projected, or data-conditioned quantity from being promoted into a stronger authority class.

The classes can coexist within one physical realization. In the compact sector, for example, the finite spherical mass state is a closed realization, the mass–spin handoff is a conditional closure, and Structural Mode, local clock, trapping, and geometric rotation quantities

are downstream post-solution readouts. Likewise, Structural Mode in resolved galaxies and pressure-supported clusters is a post-solution fingerprint of the completed field rather than an input used to produce that field.

This authority hierarchy also determines the validation sequence. The operational rule is

$$\begin{aligned} \text{derive the relation} \quad &\longrightarrow \quad \text{assign its authority class} \\ &\downarrow \quad \text{freeze the upstream specification for the declared test} \\ &\downarrow \quad \text{project or reduce as required} \\ &\longrightarrow \quad \text{compare with the independent target.} \end{aligned} \tag{19}$$

"Freeze" here is a validation rule, not a statement that CPTG is permanently unrevisable. It means that a failed downstream comparison cannot be rescued within the same declared test by retrospectively changing the upstream law, changing its authority class, or inserting a sector-specific hidden component. A revised upstream theory may of course be proposed, but it constitutes a new derivation and a new test rather than a repair of the original validation.

Because several CPTG reductions contain conspicuous rational or π-based constants, their provenance must be stated separately from their numerical success. Table 2 records whether each major upstream quantity is a native definition, an algebraic consequence, a registered reduced-law coefficient, or a frozen comparison coordinate. A fixed coefficient is not promoted to a first-principles derivation merely because it is reused successfully downstream.

The following fixed relations and registered sector coordinates retain explicit authority boundaries:

- $H_{0,\mathrm{ac}}^{\mathrm{CPTG}} = H_0^{(\pi)}\sqrt{\mathcal{G}_T}$ is the registered present-normalization projection rule used to construct the acoustic rate coordinate. The shorthand $H_{\mathrm{transported}} = H_{\mathrm{raw}}\sqrt{\mathcal{G}_T}$ refers to this normalization map, not to a point-by-point transformation of the full rate history [12].
- $\mathcal{D}_c$, $\mathcal{A}_s$, and $\mathcal{T}_\tau$ are registered deterministic CMB comparison-map gates. They act on frozen conventional likelihood coordinates and are not native CPTG matter or radiation parameters [12].
- The displayed reduced growth/lensing pair $(\mu_{\mathrm{CPTG}}, \Sigma_{\mathrm{CPTG}})$, together with the branch-derived response scale A_L^{CPTG}, defines fixed comparison-response coordinates inherited from the upstream CPTG state. They are not adjustable dark-sector functions and do not supply freedom to retune the native branch against individual datasets [12].
- J_{CPTG} and K_{CPTG} are fixed analytic scalar inputs of the nuclear source construction and are not reaction-specific fit coefficients [16].
- $\tau_7 = \ln \pi$ is the fixed integrated mass-seven abundance-transport law applied as a live source-network sink to ^{7}Li and ^{7}Be [17, 18].

Quantity	Origin / defining relation	Derivation status in this parent paper	Authority
p_C, $\mathcal{B}_\pi$, $\mathcal{T}_\pi$	$p_C = \pi$, $\mathcal{B}_\pi = 1 - 1/\pi$, $\mathcal{T}_\pi = 1/\pi$	p_C is the native curvature exponent; the two weights are exact algebraic complements	[12]
$a_{\pi,0}$	$a_{\pi,0} = \pi^{3/2} \times 10^{-10}\,\mathrm{m\,s^{-2}}$	Native universe-scale anchor; H_∞, $H_0^{(\pi)}$, and $R_{C,0}$ are derived from it	[12]
p_{ac}, $\mathcal{G}_T$	$p_{\mathrm{ac}} = 3 - \pi/100$, $\mathcal{G}_T = p_{\mathrm{ac}}/p_C$	p_{ac} is the fixed acoustic-projection exponent; $\mathcal{G}_T$ is its exact projection ratio	[12, 15]
η_{10}^{ac}	$(\Omega_b h^2)_{\mathrm{ac}} = 0.0225278574944$ with the fixed-temperature conversion $\eta_{10} \simeq 273.9\,\Omega_b h^2$	Frozen acoustic baryon-to-photon comparison coordinate entering the primordial nuclear construction	[12, 16]
$N_{\mathrm{eff}}^{\mathrm{CPTG}}$	$N_{\mathrm{eff}}^{\mathrm{CPTG}} = p_{\mathrm{ac}} = 3 - \pi/100$	Effective relativistic-degree comparison coordinate used by the primordial radiation mapping; it is not an additional native radiation species	[18]
Resolved suppression exponents	123/50 and $-11/50$ in the nonlinear suppression factor	Registered exponents of the support-conditioned resolved law; they are retained in the galaxy and pressure-supported cluster realizations and are not inferred separately for each object or radius	Registered resolved law

Table 2: Provenance and derivation status of major upstream fixed quantities. The table distinguishes native and exact algebraic quantities from registered reduced-law coefficients and frozen comparison coordinates; reduction-specific coefficients are not promoted to universal parent invariants by successful downstream use.

The registered CMB comparison-map gates named above are

$$\mathcal{D}_c = \sqrt{\mathcal{G}_T}\left(1 + \frac{\pi - 3}{30\pi}\right) = 0.9735353159758136, \tag{20}$$

$$\mathcal{A}_s = \frac{\pi}{3} = 1.0471975511965976, \tag{21}$$

$$\mathcal{T}_\tau = \mathcal{G}_T^2 = 0.8928920596100127. \tag{22}$$

These are derived comparison coordinates acting on the frozen CMB comparison row; they are not additional native cosmological variables or independently fitted amplitudes [12].

2.5 Structural Acceleration and Regime-Equivalent Scale

CPTG ties the geometric response of a gravitating system to the structure of the solved state rather than assigning one universal acceleration constant to every object. The symbol $a_\star$ is used in realizations where the structural acceleration appears explicitly; in other regimes the same structural role may be carried by the natural acceleration normalization of the solved geometry. The coordinate realization is therefore regime-dependent even though the physical requirement is common: the response scale belongs to the object and state being solved, not to an externally imposed universal acceleration.

Where $a_\star$ is used explicitly it is nonnegative,

$$a_\star \in [0, \infty), \tag{23}$$

and it must remain distinct from Structural Mode N. Structural acceleration participates in constructing or characterizing the solved gravitational state; Structural Mode is measured only afterward as a fingerprint of how that completed state is organized.

In the registered resolved galaxy and pressure-supported cluster realization, the structural acceleration is local rather than one object-wide fitted constant:

$$a_\star(r_i) = g_{\rm sup}(r_i). \tag{24}$$

Each resolved radius therefore carries its own structural acceleration. For a rotation-supported galaxy, $g_{\rm sup} = V_{\rm obs}^2/r$ at the measured point. For a pressure-supported system, $g_{\rm sup}$ is reconstructed from the declared hydrostatic or Jeans support state. No object-wide average is substituted for the resolved profile, and the structural coordinate is recalculated from the state appropriate to each radius or shell.

The same physical role need not use the symbol $a_\star$ in every regime. A stationary compact black hole is written in a mass-scaled strong-field coordinate whose natural acceleration normalization is fixed by the retained exterior gravitational mass, with spin completing the stationary compact state. The compact normalization is therefore a regime-equivalent structural scale rather than a galaxy $a_\star$ copied into the strong-field problem. Likewise, explicitly anisotropic or evolving systems may carry their structural state through time-dependent support and transport variables rather than through one universal scalar acceleration history.

In resolved weak-field applications the structural acceleration plays an acceleration-scale role analogous in dimension to the MOND constant a_0, but the assignment is fundamentally different: MOND fixes one universal a_0, whereas CPTG evaluates the structural scale from the object and state being solved [3]. The analogy is one of dynamical role and dimension, not an identification of laws, interpolation functions, or numerical values.

The universe-scale quantity $a_{\pi,0}$ introduced later is separate again. It is the native acceleration anchor of the cosmological π branch and is not obtained by assigning a local bound-system structural acceleration to the Universe. Conversely, CMB-inferred, luminosity-distance, lensing, abundance, or other observational coordinates do not become local structural accelerations merely because they can be written in acceleration or inverse-time units.

This distinction supplies the practical rule used throughout the paper: identify the physical regime and its native structural scale or equivalent first, solve the corresponding CPTG realization second, and only then construct Structural Mode or any observational comparison coordinate.

3 Bound and Dynamical Structure

3.1 Galaxy-Scale Curvature Polarization Reduction

The resolved galaxy realization uses the measured rotation state as the support coordinate and the observed SPARC gas, disk, and bulge components as the material source. The registered calculation is an effective radial reduction of the resolved disk data; it is not presented as a newly derived full axisymmetric tensor solution. The purpose of the reduction is to test whether one support-conditioned CPTG response can close the measured radial state without introducing an additional unseen matter source [1].

At each measured radius,

$$g_{\rm sup}(r_i) = \frac{V_{{\rm obs},i}^2}{r_i}, \qquad a_\star(r_i) = g_{\rm sup}(r_i). \tag{25}$$

Thus the structural acceleration used by the resolved calculation is local: every measured radius carries its own $a_\star(r_i)$. No object-wide average is substituted for the resolved support profile. In the numerical realization, a separate differentiable reconstruction of the measured support profile supplies derivatives for the curvature carrier, polarization operator, and radial-organization coordinate; the directly measured local $a_\star(r_i)$ values remain unchanged in the field equation.

The baryonic source is constructed directly from the tabulated SPARC velocity components. For a signed component velocity define

$$Q(V) = V|V|. \tag{26}$$

The retained source is then

$$V_b^2(r) = Q(V_{\rm gas}(r)) + Q(V_{\rm disk}(r)) + Q(V_{\rm bulge}(r))\,, \tag{27}$$

with

$$g_b(r) = \frac{V_b^2(r)}{r}. \tag{28}$$

The tabulated stellar velocity templates are used at their unit-Υ reference normalization. No additional fitted disk or bulge mass-to-light rescaling enters the CPTG source construction.

The support field supplies the field-plus-gradient carrier. For a genuinely spherical radial support field,

$$\mathbf{g}_{\rm sup}(r) = -g_{\rm sup}(r)\hat{\mathbf{r}}, \tag{29}$$

the contracted gradient is

$$(\partial_i g_{{\rm sup},j})(\partial_i g_{{\rm sup},j}) = [g'_{\rm sup}(r)]^2 + 2\left[\frac{g_{\rm sup}(r)}{r}\right]^2. \tag{30}$$

The corresponding reduced carrier is

$$C_C(r) = g_{\rm sup}^2(r) + \kappa_C^2 \left\{ [g'_{\rm sup}(r)]^2 + 2 \left[\frac{g_{\rm sup}(r)}{r} \right]^2 \right\}. \tag{31}$$

With spherical measure $r^2 dr$, the registered bare polarization-response operator is

$$B_{\rm pol}(r) = \frac{1}{r^2} \left[(r^2 + 2\kappa_C^2) g_{\rm sup} C_C^{-2/3} - \frac{d}{dr} \left(r^2 \kappa_C^2 g'_{\rm sup} C_C^{-2/3} \right) \right]. \tag{32}$$

For disk galaxies, Eqs. (31)–(32) are the registered effective radial operator acting on the resolved rotation state. The same contraction is a direct spherical realization in pressure-supported systems, treated separately below.

The local radial organization of the support field is recorded by

$$\Sigma_C^2(r) = \frac{1}{3} \left[1 - \frac{d \ln g_{\rm sup}(r)}{d \ln r} \right], \tag{33}$$

with the numerical admissibility rule

$$\Sigma_C^2 \geq 0. \tag{34}$$

This numerical admissibility condition constrains the differentiable support reconstruction, where necessary, to maintain nonnegative Σ_C^2. It does not impose an upper ceiling: values of Σ_C^2 greater than unity remain admissible and are not clipped.

The nonlinear high-response suppression is

$$\mathcal{S}(r) = \left[1 + \left(\frac{g_{\rm CPTG}(r)}{a_\star(r)} \right)^{123/50} \right]^{-11/50}. \tag{35}$$

The resolved support-conditioned field law is

$$g_{\rm CPTG}(r) = \frac{g_b(r) + A_C a_\star^{2/3}(r) B_{\rm pol}(r) \mathcal{S}(r)}{1 + c_C \Sigma_C^2(r)}. \tag{36}$$

The equation is implicit because $g_{\rm CPTG}$ appears in Eq. (35). The object-level quantities

$$\kappa_C, \qquad A_C, \qquad c_C \tag{37}$$

are reduced state/closure coordinates of the solved object, not universal CPTG constants. They belong to one coupled support-conditioned solution and have no independent authority to redefine the baryonic source, alter the registered response law, or tune a downstream observable separately from the solved field. Their dimensions are

$$[\kappa_C] = L, \qquad [A_C] = L^{2/3} T^{-4/3}, \qquad [c_C] = 1. \tag{38}$$

Because the support state is reconstructed from the observed dynamics, the resolved relation is a support-conditioned inverse realization. It determines whether the registered

CPTG response, the declared baryonic source, and the reconstructed support field can occupy one consistent solved state. It is not, by construction, a withheld-support forward prediction of the rotation curve from baryons alone. Numerical closure tests and catalog-level residual summaries belong to the dedicated resolved-gravity studies rather than to the parent law [10].

The resolved General Relativistic overlap is the joint zero-response boundary

$$A_C \to 0, \qquad c_C \to 0, \tag{39}$$

for which

$$g_{\mathrm{CPTG}}(r) \to g_b(r). \tag{40}$$

This is a state-space boundary of the resolved response, not a universal acceleration threshold at which a different gravity law switches on.

Structural Mode is evaluated only after this field has been solved. The galaxy organization operator remains distinct from the carrier used to construct the resolved field. For the downstream galaxy Structural-Mode readout, define

$$C_g(r) = g^2_{\mathrm{CPTG}}(r) + \kappa^2_{\mathrm{struct}} \left(\frac{dg_{\mathrm{CPTG}}}{dr} \right)^2, \qquad \kappa_{\mathrm{struct}} = 1 \text{ kpc}. \tag{41}$$

The fixed one-kiloparsec length in Eq. (41) belongs to the post-solution galaxy organization operator. It is not the object-level reduced coordinate κ_C entering Eq. (31). This separation prevents the Structural-Mode measurement from feeding back into the gravitational solve.

3.1.1 Relation to the registered weak-field scaling reductions

The measured-domain support-conditioned law above and the established analytic weak-field scaling reductions answer different questions. Equation (36) uses the resolved support state and does not, by itself, prescribe an unmeasured continuation of $a_\star(r)$ beyond that state. The BTFR scaling and the fixed-outer-source asymptotic continuation considered below therefore retain their separately registered weak-field galaxy reduction rather than extrapolating Eq. (25) outside the measured domain.

In that analytic weak-field reduction, $a_\star$ denotes the object-level structural scale of the declared scaling solution. On its positive branch the field is written

$$g(r) = g_{\mathrm{bar}}(r) + \frac{\sqrt{a_\star g_{\mathrm{bar}}(r)} + g_{\mathrm{geom}}(r)}{\left[1 + \left(g(r)/a_\star\right)^{123/50}\right]^{11/50}}. \tag{42}$$

The corresponding radial transport remnant is

$$g_{\mathrm{geom}}(r) = \frac{a_\star}{2} \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \qquad R_c = \frac{48\pi c^2}{a_\star}. \tag{43}$$

This scaling form is not substituted for the resolved support-conditioned law above. It is retained for the established intermediate BTFR scaling and fixed-outer-source far-outer continuation, whose assumptions and conclusions are stated separately below.

3.2 Galaxy Structure and the Baryonic Tully–Fisher Scaling

The baryonic Tully–Fisher relation (BTFR) is an analytic weak-field consequence of the registered CPTG galaxy scaling reduction. It is not introduced as a separate interpolation law and is not obtained by refitting the resolved support-conditioned calculation of Eq. (36). The two constructions answer different questions: the measured-domain law closes the resolved field radius by radius, whereas the BTFR reduction isolates the leading low-acceleration amplitude scaling under a declared intermediate-regime source approximation.

On the positive structural branch, suppose the galaxy has entered an interval in which

$$g_{\rm bar}(r) \ll a_\star, \qquad g(r) \ll a_\star, \tag{44}$$

so that the nonlinear suppression tends to unity. If the polarization term dominates the direct baryonic and residual-transport contributions over the scaling interval,

$$g_{\rm bar}(r),\ |g_{\rm geom}(r)| \ll \sqrt{a_\star g_{\rm bar}(r)}, \tag{45}$$

then Eq. (42) reduces at leading order to

$$g(r) \simeq \sqrt{a_\star g_{\rm bar}(r)}. \tag{46}$$

For the intermediate flat-regime scaling argument, approximate the exterior baryonic source by an effective enclosed mass,

$$g_{\rm bar}(r) \simeq \frac{GM_{\rm eff}}{r^2}, \qquad g(r) = \frac{V^2(r)}{r}. \tag{47}$$

Substitution into Eq. (46) gives

$$V_f^4 \simeq G a_\star M_{\rm eff}. \tag{48}$$

Writing the declared source mapping as

$$M_{\rm eff} = \Xi M_b, \tag{49}$$

where Ξ is the structural source-mapping factor associated with the approximation over the scaling interval and is not an added gravitating component, gives

$$V_f^4 \simeq G a_\star \Xi M_b. \tag{50}$$

The fourth-power form is therefore fixed by the polarization-dominated weak-field scaling. Its population normalization is carried by the structural product

$$A_{\rm str}(M_b) \equiv a_\star(M_b)\, \Xi(M_b). \tag{51}$$

Consequently,

$$\frac{d\ln V_f}{d\ln M_b} = \frac{1}{4}\left[1 + \frac{d\ln A_{\rm str}}{d\ln M_b}\right]. \tag{52}$$

An exact quarter-power population slope follows when the structural product has no systematic mass dependence over the population being compared. The parent result is therefore stronger than a dimensional analogy but narrower than a claim that every catalog construction must return an exactly universal normalization: CPTG fixes the fourth-power weak-field scaling and states explicitly the structural condition under which its normalization is common across a population [2].

The BTFR compresses only the leading weak-field amplitude relation. It does not encode where the solved curvature response is organized across radius. That additional information belongs to the post-solution Structural Mode fingerprint introduced next.

3.3 Structural Mode N: Cross-Regime Post-Solution Organization and CSMI

Structural Mode N is the CPTG post-solution measure of how a completed gravitational state is internally organized. Its parent meaning is

$$N = \frac{R}{\lambda}, \tag{53}$$

where R is the structural extent of the declared solved state and λ is the geometry-appropriate organization length extracted from that state. Structural Mode is therefore not a fitting parameter, not an acceleration scale, not a morphology label, and not a prospective value assigned before the gravitational problem is solved. It is an object-specific structural fingerprint read from the completed CPTG state.

The physical meaning of Eq. (53) is common across gravitational regimes, but the operator used to construct λ follows the geometry of the solution. A resolved disk galaxy, a spherical pressure-supported cluster, and a finite rotating black hole therefore do not share one copied centroid formula. What is universal is the inference order,

$$\text{solved CPTG state} \longrightarrow \text{geometry-appropriate organization measure} \longrightarrow N, \tag{54}$$

with any catalog label, population comparison, clock readout, or other interpretation downstream of the solved fingerprint. Independent reconstructions remain independent measurements; Structural Mode values are not averaged or pooled into a synthetic object fingerprint.

3.3.1 Resolved galaxy realization

For a solved radial galaxy field, the structural extent is the outer retained radius,

$$R = \max(r_i). \tag{55}$$

The downstream galaxy organization operator is the carrier already defined in Eq. (41),

$$C_g(r) = g^2_{\mathrm{CPTG}}(r) + \kappa^2_{\mathrm{struct}} \left(\frac{dg_{\mathrm{CPTG}}}{dr} \right)^2, \qquad \kappa_{\mathrm{struct}} = 1 \text{ kpc}, \tag{56}$$

with structural weight

$$w_{N,\mathrm{gal}}(r) = \left| \frac{dC_g}{dr} \right|. \tag{57}$$

The registered outer-field organization length is

$$\lambda_{N,\mathrm{gal}} = \frac{\displaystyle\int_{R/5}^{R} r\, w_{N,\mathrm{gal}}(r)\, dr}{\displaystyle\int_{R/5}^{R} w_{N,\mathrm{gal}}(r)\, dr}, \qquad N_{\mathrm{gal}} = \frac{R}{\lambda_{N,\mathrm{gal}}}. \tag{58}$$

The interval $[R/5, R]$ and the fixed $\kappa_{\mathrm{struct}} = 1\,\mathrm{kpc}$ are registered features of the galaxy readout. They are applied only after the registered resolved field has closed and therefore do not participate in the resolved gravitational solve. Whenever the structural weight is not identically zero on the retained interval,

$$\frac{R}{5} \le \lambda_{N,\mathrm{gal}} \le R, \qquad 1 \le N_{\mathrm{gal}} \le 5. \tag{59}$$

If the structural weight vanishes identically on the retained window, the centroid denominator vanishes and N_{gal} is undefined rather than assigned an arbitrary value.

The galaxy realization retains the useful amplitude invariance of the Structural-Mode construction. Under

$$C_g(r) \longrightarrow A\, C_g(r) + B, \qquad A \neq 0, \tag{60}$$

the common factor $|A|$ cancels from the centroid. Likewise, if

$$\widetilde{r} = sr, \qquad \widetilde{R} = sR, \qquad \widetilde{C}_g(\widetilde{r}) = AC_g(\widetilde{r}/s) + B, \tag{61}$$

with $s > 0$ and $A \neq 0$, then $\widetilde{\lambda} = s\lambda$ and $\widetilde{N} = N$. Because the galaxy operator contains the fixed physical scale $\kappa_{\mathrm{struct}} = 1\,\mathrm{kpc}$, arbitrary physical resizing of a galaxy does not automatically satisfy this conditional similarity map. The invariant content is therefore organization of the solved field, not merely its amplitude or an arbitrary rescaling of radius.

3.3.2 Spherical pressure-supported cluster realization

A galaxy cluster is not assigned Structural Mode by copying the disk-galaxy operator. For a solved spherical pressure-supported field $g_C(r)$, the three-dimensional organization carrier is

$$C_{N,C}(r) = 3g^2_C(r) + \left[r \frac{dg_C}{dr} \right]^2, \qquad w_{N,C}(r) = \left| \frac{dC_{N,C}}{dr} \right|. \tag{62}$$

For the outer retained radius R of that reconstruction, the spherical shell measure gives

$$\lambda_{N,C} = \frac{\int_{R/5}^{R} r^3 w_{N,C}(r)\, dr}{\int_{R/5}^{R} r^2 w_{N,C}(r)\, dr}, \qquad N_C = \frac{R}{\lambda_{N,C}}. \tag{63}$$

The factor of three and the shell weighting arise from the spherical organization of the solved radial field. Independent reconstructions of the same cluster retain separate N_C fingerprints, each tied to its own solved support field and retained radial domain. Such fingerprints are not averaged into a synthetic cluster mode. The cluster gravitational reduction is stated below; numerical applications belong to the dedicated cluster studies.

3.3.3 Finite compact realization

The same parent definition also extends to finite black-hole states, but the organization measure must follow the compact relativistic geometry. Let

$$x = \frac{r}{r_m}, \qquad q = |\hat{a}_\infty|, \tag{64}$$

where $r_m = GM_\infty/c^2$ is the compact matching scale and q is the magnitude of the exterior dimensionless spin. The Kerr-compatible equal-measure radial coordinate used by the registered compact readout is

$$\xi(x, q) = \left[\frac{x(x^2 + q^2)}{1 + q^2}\right]^{1/3}. \tag{65}$$

If $w_{N,\mathrm{BH}}(x)$ denotes the structural weight extracted from the closed finite compact state, the organization length and Structural Mode are

$$\lambda_{N,\mathrm{BH}} = r_m \frac{\int_0^1 \xi(x, q) w_{N,\mathrm{BH}}(x)\, dx}{\int_0^1 w_{N,\mathrm{BH}}(x)\, dx}, \tag{66}$$

$$N_{\mathrm{BH}}(q) = \frac{r_m}{\lambda_{N,\mathrm{BH}}} = \frac{\int_0^1 w_{N,\mathrm{BH}}(x)\, dx}{\int_0^1 \xi(x, q) w_{N,\mathrm{BH}}(x)\, dx}. \tag{67}$$

The compact structural weight is not an independently fitted function; it is obtained from the finite mass–spin state described in the compact section below. Mass and r_m cancel from the dimensionless ratio, so the registered rotating fingerprint is fixed by the solved compact geometry rather than by an additional structural fit. The compact realization subsequently uses the measured mode to identify an organization shell carrying definite clock and frame-dragging readouts; those strong-field quantities remain downstream of the closed compact state.

Regime	Solved state	Organization measure	Structural fingerprint
Galaxy	Resolved radial CPTG field	Outer-field centroid of $\lvert dC_g/dr\rvert$ on $R/5 \le r \le R$	$N_{\rm gal} = R/\lambda_{N,\rm gal}$
Galaxy cluster	Spherical pressure-supported CPTG field	Three-dimensional shell-weighted centroid of $\lvert dC_{N,C}/dr\rvert$	$N_C = R/\lambda_{N,C}$
Black hole	Closed finite mass–spin compact state	Kerr-compatible equal-measure coordinate ξ with compact structural weight	$N_{\rm BH} = r_m/\lambda_{N,\rm BH}$

Table 3: Cross-regime Structural Mode architecture. The common physical definition is $N = R/\lambda$; the organization operator follows the geometry of the solved state rather than being copied between object classes.

3.3.4 CSMI and downstream structural interpretation

The Curvature Structure Mode Index (CSMI) is downstream of the continuous Structural Mode fingerprint. In its galaxy realization, CSMI assigns descriptive labels to recurring intervals of the solved $N_{\rm gal}$ population without feeding Hubble type, bulge prominence, surface-brightness class, or any other morphology label into the gravitational solve. The continuous value of N has priority over any particular catalog partition. CSMI is a galaxy catalog representation; the parent Structural Mode concept extends across gravitational geometries without imposing one catalog partition on clusters or compact objects [8].

No universal cross-object CSMI is imposed on clusters or black holes in this parent paper. Those objects carry their own solved Structural Mode fingerprints. Any object-class catalog constructed for those regimes is downstream of the individual fingerprints and remains separate from the gravitational solution itself.

The parent inference direction is therefore

$$\text{solved CPTG state} \longrightarrow N \longrightarrow \text{downstream structural interpretation}, \tag{68}$$

with no reverse path by which a catalog class or desired mode value selects the gravitational solution.

3.4 Far-Outer Continuation Law

The far-outer law is a separate asymptotic consequence of the registered fixed-outer-source continuation. It is not obtained by inventing an extrapolation of the local support coordinate in Eq. (25). Instead, once the outer baryonic velocity contribution of the declared weak-field continuation is held at its terminal finite value,

$$V^2_{\rm bar,CPTG}(r) \to \text{constant}, \tag{69}$$

its effective baryonic acceleration obeys

$$g_{\rm bar}(r) = \frac{V^2_{\rm bar,CPTG}}{r} \propto r^{-1}. \tag{70}$$

On the same positive structural branch, the low-acceleration polarization response therefore scales as

$$\sqrt{a_\star g_{\rm bar}(r)} \propto r^{-1/2}. \tag{71}$$

The radial transport remnant has the asymptotic form

$$g_{\rm geom}(r) \propto r^{-1}, \tag{72}$$

and the direct source term also falls as r^{-1}. The polarization term consequently dominates the leading far-outer response,

$$g(r) \propto r^{-1/2}. \tag{73}$$

Since $V^2 = gr$, the asymptotic rotation law is

$$V(r) \propto r^{1/4}, \qquad s_\infty \equiv \frac{d\ln V}{d\ln r} = \frac{1}{4}. \tag{74}$$

Numerical convergence studies of this asymptotic law are reported separately; the parent paper retains only the analytic continuation and its limiting structure.

The asymptotic field also fixes the leading rate of approach. Writing the dimensionless outer radius as $x = r/R_{\rm max}$, the first two positive contributions have the form

$$g(x) = Px^{-1/2} + Qx^{-1} + O\left(x^{-3/2}\right), \qquad P > 0, \tag{75}$$

where P is the leading polarization amplitude and Q collects the direct-source and leading transport-remnant amplitudes. It follows that

$$V(x) = \sqrt{R_{\rm max}P}\, x^{1/4}\left[1 + \frac{Q}{2P}x^{-1/2} + O\left(x^{-1}\right)\right], \tag{76}$$

and hence

$$s(x) = \frac{1}{4} - \frac{Q}{4P}x^{-1/2} + O\left(x^{-1}\right). \tag{77}$$

Thus the same reduction predicts both the quarter-power asymptote and a leading $x^{-1/2}$ convergence correction. The approach exponent $1/2$ follows from the asymptotic expansion itself rather than from a fitted convergence curve.

The BTFR and far-outer results are therefore not interchangeable uses of one source assumption. The BTFR reduction uses an effective finite enclosed mass, $g_{\rm bar} \propto r^{-2}$, whereas the fixed-outer-source continuation gives $g_{\rm bar} \propto r^{-1}$. They are separate consequences of the registered weak-field CPTG galaxy response under two explicitly different source limits.

3.5 Pressure-Supported Bound Systems and Coherent Galaxy Clusters

A pressure-supported galaxy or coherent galaxy cluster is an extended bound-system realization of the same resolved CPTG law used for rotational support. The material source remains the physically present baryonic distribution. What changes is the observational interface used to reconstruct the support field required by the state. Ordered rotation supplies that field through V^2/r; collisionless pressure support supplies it through Jeans balance; hot gas supplies it through hydrostatic balance. These are support coordinates of one gravitational problem, not additional sources of gravity and not separate gravity laws.

For a spherical collisionless tracer population with density $\rho_\star(r)$, radial velocity dispersion $\sigma_r(r)$, and orbital-anisotropy parameter $\beta_J(r)$, the support field is

$$g_{\rm sup}(r) = -\frac{\sigma_r^2(r)}{r}\left[\frac{d\ln\rho_\star}{d\ln r} + \frac{d\ln\sigma_r^2}{d\ln r} + 2\beta_J(r)\right]. \tag{78}$$

For a hydrostatic gas with pressure $P(r)$ and mass density $\rho_g(r)$, the corresponding support field is

$$g_{\rm sup}(r) = -\frac{1}{\rho_g(r)}\frac{dP(r)}{dr}. \tag{79}$$

Equations (78) and (79) are observational dynamical interfaces. They reconstruct the acceleration required by the measured support state; they are not additional CPTG field equations. In a mixed-support system, the complete declared dynamical equilibrium supplies $g_{\rm sup}(r)$ before the CPTG response is evaluated. A one-radius proxy is therefore not promoted to the parent law when a resolved support profile is available.

The local structural acceleration in the resolved pressure-supported realization is

$$a_\star(r) = g_{\rm sup}(r), \tag{80}$$

with no object-wide average substituted for the resolved profile. A useful spherical support-equivalent mass is

$$M_{\rm sup}(r) = \frac{r^2 g_{\rm sup}(r)}{G}. \tag{81}$$

Its local-to-enclosed density ratio satisfies

$$\frac{\rho_{\rm sup}}{\bar\rho_{\rm sup}} = \frac{1}{3}\left[2 + \frac{d\ln g_{\rm sup}}{d\ln r}\right], \tag{82}$$

so the resolved support coordinate becomes

$$\Sigma_C^2(r) = 1 - \frac{\rho_{\rm sup}}{\bar\rho_{\rm sup}} = \frac{1}{3}\left[1 - \frac{d\ln g_{\rm sup}}{d\ln r}\right]. \tag{83}$$

The baryonic material source is constructed independently from the enclosed physical

baryonic mass,

$$M_b(r) = M_{\rm gas}(r) + M_\star(r) + \cdots, \qquad g_b(r) = \frac{GM_b(r)}{r^2}, \tag{84}$$

where the ellipsis denotes any additional physically present baryonic component included by the declared source model, not an unseen matter term.

For a genuinely spherical support field, the carrier in Eq. (31), the polarization operator in Eq. (32), the suppression law in Eq. (35), and the support-conditioned field law in Eq. (36) apply directly with $g_{\rm sup}(r)$ reconstructed from the pressure-support interface. The factor

$$(\partial_i g_{{\rm sup},j})(\partial_i g_{{\rm sup},j}) = [g'_{\rm sup}(r)]^2 + 2\left[\frac{g_{\rm sup}(r)}{r}\right]^2 \tag{85}$$

is then the exact spherical contraction rather than an effective radial disk representation.

The coherent pressure-supported reduction therefore has the parent ordering

$$\begin{gathered}\text{baryonic source + measured support state} \longrightarrow g_{\rm sup}(r),\ a_\star(r),\ \Sigma_C^2(r) \\ \longrightarrow \text{resolved CPTG field.}\end{gathered} \tag{86}$$

Structural Mode is measured only after that field has been solved, using the spherical operator of Eq. (63). Independent support reconstructions remain independent solved states and therefore retain independent Structural Mode fingerprints.

Cosmological redshift does not define a multiplicative rescaling of the local structural acceleration or supply a separate bound-system gravity law. Any cosmology-dependent conversion used to construct an observational source or support coordinate remains downstream of the local gravitational reduction rather than redefining it.

3.6 Anisotropic Dynamical Curvature Transport

The coherent bound-system reductions above compress the transport response into a radial or spherical solved state. That reduction is not appropriate when the source geometry and dynamical evolution are strongly anisotropic. In a dissociative or otherwise directionally separating system, CPTG retains the transport direction explicitly. Curvature polarization continues to describe the nonlinear local response to the ordinary material source, while curvature transport describes the persistence and redistribution of the organized geometric response through the evolving system.

For a declared organized transport direction S^α, define the reduced transport scalar

$$\Psi_{\rm tr} = S^\alpha \Sigma_{\rm tr} \nabla_\alpha \Phi, \tag{87}$$

so that the directional transport representative is

$$\mathcal{L}^{\rm (red)}_{\rm trans} = -\frac{\lambda_t}{2}\Psi^2_{\rm tr}. \tag{88}$$

Here S^α is the geometry-selected transport direction, $\Sigma_{\rm tr}$ is the transport-retention or suppression coordinate of the declared reduction, and λ_t is the transport coupling. Holding the selected direction fixed only within that reduced variation, the leading directional contribution obtained by varying Φ is

$$\mathcal{O}_{\rm trans}[\Phi] = \nabla_\alpha\Big(\lambda_t S^\alpha S^\beta \Sigma_{\rm tr}^2 \nabla_\beta \Phi\Big)\,, \tag{89}$$

with the additional nonlinear contribution from $\delta\Sigma_{\rm tr}$ belonging to the complete reduced variation whenever the transport-retention coordinate depends on the solved field. The directional tensor $S^\alpha S^\beta$ carries the anisotropic organization of transport; it is not an additional stress–energy source.

The ordinary material source remains whatever physically present baryonic components participate in the merger or dynamical system. For a gas-plus-stellar decomposition one may write

$$\Phi_{\rm bary} = \Phi_{\rm gas} + \Phi_{\rm star}, \tag{90}$$

where both terms are ordinary source contributions. Spatial separation between those components changes the source geometry supplied to the CPTG response; it does not license an additional unseen matter distribution.

For a reduced spatial projection onto a dominant direction $\mathbf{S}$, the leading anisotropic transport interface takes the form

$$\mathcal{K}_{\rm aniso} = \nabla^2\Phi - s_{\rm tr}\,\nabla\cdot\Big[\Sigma_{\rm tr}^2\,\mathbf{S}\,(\mathbf{S}\cdot\nabla)\Phi\Big]\,, \tag{91}$$

where $s_{\rm tr}$ collects the coupling and normalization appropriate to the declared spatial reduction. Equation (91) is a projection of the directional transport structure of Eq. (89); it is not a replacement for the parent covariant field equation and does not turn a projected observable into a new matter source.

The anisotropic hierarchy is therefore

$$\begin{aligned} &\text{ordinary material source + dynamical geometry} \longrightarrow \text{curvature polarization} \\ \longrightarrow\ &\text{directional curvature transport} \longrightarrow \text{solved anisotropic gravitational state} \\ &\longrightarrow \text{observational projection.} \end{aligned} \tag{92}$$

This ordering separates the physical transport law from any particular lensing, merger-plane, or image reconstruction. In the common limit where explicit directional transport becomes negligible or the dynamical state closes back to an approximately radial support geometry, the anisotropic reduction must agree with the corresponding coherent bound-system description according to Eq. (2). If the full polarization–transport response also becomes negligible, the same state approaches the GR limit of Eq. (9).

3.7 Finite Compact Gravity: Spherical State and Mass Organization

Compact black holes occupy the strong-field end of the same CPTG source–polarization–transport hierarchy. General Relativity remains the controlling exterior description wherever the additional CPTG response is negligible, including the ordinary Schwarzschild or Kerr exterior and its horizon structure. CPTG differs in the completed interior continuation: an infinite-curvature singular endpoint is not admitted as a physical completed state. The compact sector therefore requires a finite interior that joins the GR-controlled exterior continuously.

The closed compact construction begins in spherical symmetry. Let

$$x = \frac{r}{r_m}, \qquad 0 \le x \le 1, \tag{93}$$

where the finite-to-GR matching scale is

$$r_m = \frac{GM_\infty}{c^2}. \tag{94}$$

Here M_∞ is the conserved exterior gravitational mass parameter measured by the asymptotic Schwarzschild or Kerr solution. The registered minimum analytic finite profile is

$$F(x) = \frac{1}{8}\left(63x^5 - 90x^7 + 35x^9\right). \tag{95}$$

At the outer edge,

$$F(1) = 1, \qquad F'(1) = 0, \qquad F''(1) = 0. \tag{96}$$

These are matching conditions rather than adjustable shape parameters. Within the retained minimum-degree analytic odd-polynomial realization they fix Eq. (95) uniquely. They are not asserted as a uniqueness theorem over every finite compact geometry obtainable from the parent action.

The same profile is regular at the center:

$$F(x) = \frac{63}{8}x^5 + \mathcal{O}(x^7), \qquad x \to 0. \tag{97}$$

Accordingly, the mass-scaled radial carrier

$$U(x) \equiv \frac{F(x)}{x^2} = \frac{63}{8}x^3 + \mathcal{O}(x^5) \longrightarrow 0 \tag{98}$$

softens toward the center rather than developing a point-mass-like divergence. The matching surface is therefore a geometric handoff between a finite interior and the retained exterior; it is not defined as a universal polarization–transport activation radius and is not, by definition, identical to an event horizon or to any internal organization or trapping surface.

3.7.1 Misner–Sharp compact state

For the spherical compact state, CPTG uses the Misner–Sharp mass as the covariant enclosed gravitational mass. Define

$$\mathcal{F}[g] = \frac{m_{\mathrm{MS}}[g]}{M_\infty} = \frac{c^2 r}{2GM_\infty}\left(1 - D^a r D_a r\right). \tag{99}$$

For the registered minimum realization, $\mathcal{F} = F(x)$. The exterior charge is decomposed as

$$M_\infty = M_{\mathrm{c}} + M_{\mathrm{geom}}, \tag{100}$$

where M_{c} is the localized physical compact-source contribution of the stationary post-formation state and M_{geom} is the completed geometric mass-equivalent contribution. M_{c} is distinct from a progenitor mass or pre-collapse stellar mass; formation history determines which stationary compact state is reached, while Eq. (100) describes that completed state.

The minimum no-extra-shape source–transport–polarization organization is

$$\mathscr{M}_{\mathrm{c}}[g] = M_{\mathrm{c}}\mathcal{F}, \tag{101}$$

$$\mathscr{M}_{\mathrm{tr}}[g] = M_{\mathrm{geom}}\mathcal{F}(1 - \mathcal{F}), \tag{102}$$

$$\mathscr{M}_{\mathrm{pol}}[g] = M_{\mathrm{geom}}\mathcal{F}^2. \tag{103}$$

The three channels close identically:

$$\mathscr{M}_{\mathrm{c}} + \mathscr{M}_{\mathrm{tr}} + \mathscr{M}_{\mathrm{pol}} = M_\infty\mathcal{F} = m_{\mathrm{MS}}. \tag{104}$$

The source channel follows the localized physical compact matter. The transport channel carries finite interior redistribution through the transition, while the polarization channel accumulates the completed geometric contribution to the exterior gravitational charge. At the matching surface $\mathcal{F} = 1$,

$$\mathscr{M}_{\mathrm{c}}(r_m) = M_{\mathrm{c}}, \qquad \mathscr{M}_{\mathrm{tr}}(r_m) = 0, \qquad \mathscr{M}_{\mathrm{pol}}(r_m) = M_{\mathrm{geom}}. \tag{105}$$

Thus the local transport channel has completed at the handoff, while the integrated geometric contribution remains part of the conserved exterior mass. Together with $F'(1) = F''(1) = 0$, this permits the finite interior to join an ordinary GR exterior without requiring the completed geometric contribution to vanish.

3.7.2 Conditional mass–spin handoff

Rotation is introduced after the spherical mass state has closed. The handoff preserves the exterior gravitational charge and the same total angular momentum J while separating the physical compact-source mass from the completed geometric contribution. Define

$$A_{\mathrm{pol}} \equiv \frac{M_\infty}{M_{\mathrm{c}}}, \qquad M_\infty = A_{\mathrm{pol}} M_{\mathrm{c}}, \tag{106}$$

and the dimensionless exterior and compact-source spins

$$\hat{a}_\infty = \frac{cJ}{GM_\infty^2}, \qquad \hat{a}_{\rm c} = \frac{cJ}{GM_{\rm c}^2}. \tag{107}$$

The retained finite-to-Kerr handoff requires the same matching radius on the exterior and compact-source sides,

$$r_m = \frac{GM_\infty}{c^2} = \frac{GM_{\rm c}}{c^2}\left(1 + \sqrt{1 - \hat{a}_{\rm c}^2}\right). \tag{108}$$

Hence

$$A_{\rm pol} = 1 + \sqrt{1 - \hat{a}_{\rm c}^2}, \qquad 1 \le A_{\rm pol} \le 2. \tag{109}$$

Conservation of J gives

$$\hat{a}_\infty = \frac{\hat{a}_{\rm c}}{A_{\rm pol}^2}, \qquad \hat{a}_{\rm c} = A_{\rm pol}^2 \hat{a}_\infty. \tag{110}$$

Eliminating $\hat{a}_{\rm c}$ yields the exterior-state closure

$$\hat{a}_\infty^2 A_{\rm pol}^3 + A_{\rm pol} - 2 = 0. \tag{111}$$

For $0 \le |\hat{a}_\infty| \le 1$, the physical branch is the root with $1 \le A_{\rm pol} \le 2$. The compact-source and geometric contributions are therefore

$$M_{\rm c} = \frac{M_\infty}{A_{\rm pol}}, \qquad M_{\rm geom} = M_\infty - M_{\rm c}. \tag{112}$$

The branch endpoints give

$$\frac{1}{2} \le \frac{M_{\rm c}}{M_\infty} \le 1, \qquad 0 \le \frac{M_{\rm geom}}{M_\infty} \le \frac{1}{2}. \tag{113}$$

Every nonextremal state on this registered branch therefore has $M_{\rm c} < M_\infty$: part of the exterior gravitational charge is carried by the completed geometry rather than by an additional material source. For fixed exterior gravitational charge M_∞, a positive $M_{\rm geom}$ means that the compact object can contain less localized physical source mass than the exterior mass parameter alone would imply.

Writing $q \equiv |\hat{a}_\infty|$, the retained Kerr exterior has

$$r_+^{\rm Kerr} = r_m\left(1 + \sqrt{1 - q^2}\right), \tag{114}$$

so

$$r_m < r_+^{\rm Kerr} \quad (0 \le q < 1), \qquad r_m = r_+^{\rm Kerr} \quad (q = 1). \tag{115}$$

The finite-to-Kerr handoff therefore lies inside the ordinary outer horizon throughout the subextremal branch and coincides with it only at exact extremality. The horizon is retained rather than replaced.

The authority of this construction is layered. The Misner–Sharp source–transport–polarization state is the closed spherical compact continuation. The mass–spin relation above is a conditional handoff that preserves the stated exterior charges. Kerr-compatible Structural Mode, trapping, clock, and frame-dragging quantities are downstream post-solution readouts and are introduced separately; they do not replace the spherical mass law with a new fitted axisymmetric interior equation.

3.7.3 Compact organization, trapping, time, and geometric rotation

The closed spherical compact state and conditional mass–spin handoff also determine a registered set of downstream geometric readouts. These quantities do not select the compact state and do not introduce a second rotating mass law. They measure the organization and local strong-field geometry of the state after the spherical closure and mass–spin handoff have been fixed.

The compact Structural Mode defined in Eq. (67) obtains its structural weight directly from the finite profile. With

$$U(x) = \frac{F(x)}{x^2}, \tag{116}$$

define

$$\mathcal{C}_{\rm BH}(x) = U(x)^2 + [U'(x)]^2 + 2\left[\frac{U(x)}{x}\right]^2, \qquad w_{N,\rm BH}(x) = \left|\frac{d\mathcal{C}_{\rm BH}}{dx}\right|. \tag{117}$$

Substitution of Eq. (117) into Eq. (67) fixes the compact Structural Mode fingerprint once the mass–spin state is known. The corresponding organization shell is the unique retained solution of

$$x_N\left(x_N^2 + q^2\right) = \frac{1+q^2}{N_{\rm BH}^3}, \qquad r_N = x_N r_m, \tag{118}$$

with $q = |\hat{a}_\infty|$. Structural Mode therefore identifies a definite internal location in the registered compact geometry rather than serving only as a scalar label.

The same Kerr-compatible post-solution readout defines

$$\delta(x,q) = x^2 - 2xF(x) + q^2, \tag{119}$$

$$\sigma(x,\theta,q) = x^2 + q^2\cos^2\theta, \tag{120}$$

$$\mathcal{A}_d(x,\theta,q) = (x^2+q^2)^2 - q^2\delta(x,q)\sin^2\theta. \tag{121}$$

For the subextremal branch $0 \le q < 1$, the registered inner trapping surface is the physical root

$$\delta(x_-,q) = 0, \qquad 0 < x_- < 1, \qquad r_-^{\rm CPTG} = x_- r_m. \tag{122}$$

At exact extremality,

$$q = 1: \qquad \delta(1,1) = 0, \qquad x_- = 1. \tag{123}$$

Thus the inner trapping surface, the Structural-Mode organization shell, and the finite-to-Kerr matching surface are separately defined geometric quantities. The theory does not identify them by assumption.

For the locally nonrotating geometry, the lapse is

$$\alpha(x,\theta;q) = \sqrt{\frac{\delta\,\sigma}{\mathcal{A}_d}}, \tag{124}$$

with determinant-measure angular average

$$\bar{\alpha}(x;q) = \frac{\displaystyle\int_0^\pi \alpha(x,\theta;q)\,\sigma(x,\theta,q)\sin\theta\,d\theta}{\displaystyle\int_0^\pi \sigma(x,\theta,q)\sin\theta\,d\theta}. \tag{125}$$

The Structural-Mode clock is therefore

$$T_{24,N}(q) = 24\,\bar{\alpha}(x_N(q);q)\ \ \text{h}. \tag{126}$$

It reports the proper-time accumulation at the organization shell during 24 source-local asymptotic exterior-Kerr hours. It is a geometric readout at the shell selected by Structural Mode, not an independent clock used to construct the compact solution. Close to a simple inner trapping root, with $d = r_-^{\mathrm{CPTG}} - r > 0$, the local proper-time accumulation has the leading behavior

$$T_{24}(d) \propto \sqrt{\frac{d}{r_m}}. \tag{127}$$

The same registered geometry fixes the local zero-angular-momentum-frame dragging magnitude,

$$\omega(x,\theta;q) = \frac{c}{r_m}\frac{2q\,xF(x)}{\mathcal{A}_d(x,\theta,q)}, \tag{128}$$

and its determinant-measure angular average

$$\bar{\omega}(x;q) = \frac{\displaystyle\int_0^\pi \omega(x,\theta;q)\,\sigma(x,\theta,q)\sin\theta\,d\theta}{\displaystyle\int_0^\pi \sigma(x,\theta,q)\sin\theta\,d\theta}. \tag{129}$$

At the organization shell the dimensional rate separates from the dimensionless geometry as

$$\bar{\omega}_N = \frac{c}{r_m}\,\hat{\omega}_N(q). \tag{130}$$

This is geometric frame dragging of a locally nonrotating frame. It is not a material rotation rate, a fluid angular velocity, or a rigid-body spin law.

3.7.4 Stationary strong-field consequences

The compact law fixes several consequences without importing an additional phenomenological interior. First, the completed physical state is finite: an infinite-curvature singular continuation is excluded from the CPTG solution class. Second, the ordinary Schwarzschild or Kerr exterior and outer horizon remain intact wherever the polarization–transport response has completed. The finite interior therefore does not require a horizonless material surface outside the would-be GR horizon.

Because the exterior geometry is retained, stationary observables determined entirely by that exterior remain on the corresponding GR branch for the same conserved exterior charges. A non-GR ringdown shift, tidal response, or secondary-wave feature cannot be inserted solely because the interior is finite; it must follow from perturbations of the CPTG response sectors or from nonstationary dynamics derived on the solved compact background. Likewise, a generic reflective-surface echo is not implied by the stationary construction, because the finite-to-Kerr handoff is an internal geometric boundary rather than a material wall replacing the horizon [11].

The ordinary exterior horizon also means that finite interior structure does not by itself remove the semiclassical Hawking mechanism. The evaporation history and endpoint require a separate semiclassical evolution law, but any such evolution must act on the finite source–geometry system rather than terminate in the excluded infinite singular continuation [11].

Formation history is likewise distinct from the stationary compact law. Stellar collapse, fallback, binary interaction, compact-object mergers, and primordial formation determine how a system reaches its exterior charges and matter state; the equations above determine the admissible stationary continuation once those boundary data are fixed. Primordial black holes are therefore permitted by the same finite compact solution class, while their abundance, survival, and formation spectrum belong to the native cosmological sector rather than to the stationary compact equations.

The compact authority hierarchy is consequently

$$\begin{aligned} &\text{spherical finite mass state} \longrightarrow \text{conditional mass–spin handoff} \\ &\qquad\qquad\qquad\qquad \longrightarrow \text{Kerr-compatible post-solution readouts.} \end{aligned} \tag{131}$$

The final stage contains Structural Mode, organization shell, trapping surface, local clock, and frame dragging. It is a registered readout of the already closed compact state and is not promoted here to a separately solved general axisymmetric interior field equation.

4 Native Cosmology

4.1 The Native Cosmological π Branch

The structural reductions developed above are local: a galaxy, compact object, merger, or pressure-supported system is solved from the baryonic source and the geometric state appropriate to that system. Cosmology requires a different starting point. The background itself must be specified before any observational channel can be projected from it. CPTG therefore begins its cosmological construction with one native geometric history, the π-branch [12]. Acoustic, luminosity, CMB, abundance, and other comparison quantities are introduced only after this native state has been fixed.

This ordering is essential. The native branch is not obtained by averaging observational Hubble determinations, by importing a matter inventory from another cosmological framework, or by fitting separate expansion laws to different datasets. It is the universe-scale geometric reduction of CPTG. Friedmann equations, a dark-matter component, and a cosmological-constant term are not used here as the governing definition of the branch. Conventional cosmological quantities may be used later as comparison coordinates, but they do not define the native CPTG background.

4.1.1 Geometric Partition and Native Expansion Law

The native branch is organized by the curvature exponent

$$p_C = \pi, \tag{132}$$

and by the complementary geometric weights

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \qquad \mathcal{T}_\pi = \frac{1}{\pi}, \qquad \mathcal{B}_\pi + \mathcal{T}_\pi = 1. \tag{133}$$

These quantities are geometric branch weights. They are not matter-density or dark-energy fractions.

The normalized native expansion function is

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi} = 1 - \frac{1}{\pi} + \frac{1}{\pi} a^{-\pi}. \tag{134}$$

Because the weights sum to unity,

$$E_\pi(1) = 1, \tag{135}$$

so the present phase $a = 1$ is normalized without introducing an additional present-epoch fit parameter into the branch function itself.

The native cosmological rate is then

$$H_\pi(a) = H_0^{(\pi)} E_\pi(a). \tag{136}$$

At large scale factor the transport term decays and the branch approaches

$$H_\pi(a \to \infty) = H_0^{(\pi)}\sqrt{\mathcal{B}_\pi} \equiv H_\infty. \tag{137}$$

At small scale factor the $a^{-\pi}$ term dominates. The same equation therefore contains both the earlier transport-dominated regime and the later geometric asymptote; they are limits of one native history rather than separately normalized cosmologies.

In redshift form, with $a = (1+z)^{-1}$,

$$E_\pi(z)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi(1+z)^\pi, \qquad H_\pi(z) = H_0^{(\pi)}E_\pi(z). \tag{138}$$

4.1.2 Acceleration Anchor and Present Normalization

The universe-scale branch carries the native defining acceleration anchor

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10}\ \mathrm{m\,s^{-2}} = 5.568327996831708 \times 10^{-10}\ \mathrm{m\,s^{-2}}. \tag{139}$$

The dimensionful normalization in Eq. (139) is a defining native branch anchor: $\pi^{3/2}$ fixes its dimensionless geometric coefficient, while the displayed $10^{-10}\,\mathrm{m\,s^{-2}}$ sets the declared universe-scale dimensional normalization. Quantities such as H_∞, $H_0^{(\pi)}$, and $R_{C,0}$ are then derived from that anchor rather than used to fit it.

This is not the structural acceleration scale $a_\star$ of a galaxy or compact object. The two quantities belong to different geometric states and must not be interchanged.

For the native cosmological branch, the anchor fixes the asymptotic rate through

$$H_\infty = \frac{a_{\pi,0}}{c} = 57.3131992073\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{140}$$

and therefore

$$H_0^{(\pi)} = \frac{H_\infty}{\sqrt{\mathcal{B}_\pi}} = 69.4162507897\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{141}$$

The dependency chain is consequently

$$a_{\pi,0} \longrightarrow H_\infty \longrightarrow H_0^{(\pi)} \longrightarrow H_\pi(a), \tag{142}$$

not an interpolation between external CMB and distance-ladder determinations.

Equation (140) is specific to the universe-scale native anchor. It does not identify a bound-system structural acceleration with a cosmological Hubble rate. Any rate-like coordinate constructed from an object's own structural acceleration belongs to the bound-system/native-phase comparison interface rather than to the native cosmological branch. In particular, selecting a different native phase does not by itself rescale the object's local source, support state, structural acceleration, solved field, or Structural Mode.

4.1.3 Native Curvature-Transport Scale

The corresponding present native curvature-transport radius is

$$R_{C,0} = 48\pi \frac{c^2}{a_{\pi,0}} = 788.7823091477 \text{ Gpc}. \tag{143}$$

Here the capitalized R_C denotes the cosmological native curvature–transport scale. It is distinct from any bound-system structural curvature coordinate or visible object radius, and it is not an FLRW spatial-curvature radius, a photon mean free path, or a dark-matter scale.

To keep the local and universe-scale acceleration coordinates visually distinct, define the epoch-dependent native cosmological acceleration coordinate

$$a_\pi(a) \equiv a_{\pi,0} E_\pi(a). \tag{144}$$

The native radius then evolves as

$$R_C(a) = \frac{48\pi c^2}{a_\pi(a)} = \frac{R_{C,0}}{E_\pi(a)}. \tag{145}$$

The rate and radius are reciprocal expressions of the same branch state:

$$H_\pi(a) R_C(a) = H_0^{(\pi)} R_{C,0} = \frac{48\pi c}{\sqrt{\mathcal{B}_\pi}}. \tag{146}$$

Thus increasing native rate corresponds to decreasing native curvature–transport radius along the same geometric trajectory. Equation (146) is a native geometric relation and not an FLRW horizon substitution. No second background history is required to relate the two.

4.1.4 Geometric Phase and Epoch Labels

For the parent paper the common cosmological phase is written

$$N_\pi \equiv \ln a. \tag{147}$$

Dedicated cosmology papers often write this phase simply as $N = \ln a$; the subscript is retained here to avoid collision with the established galaxy Structural Mode N. Equal N_π means equal position on the native geometric trajectory.

Redshift and radiation temperature may label that same trajectory through the ordinary epoch relations

$$a = \frac{1}{1+z}, \qquad \frac{T}{T_0} = a^{-1} = 1 + z. \tag{148}$$

The temperature relation is an epoch translation, not a new CPTG state variable, not a replacement time coordinate, and not an independent thermal branch. Once the native

trajectory is specified, z or T/T_0 simply selects the corresponding value of a, and therefore of E_π, H_π, a_π, and R_C.

4.1.5 Kinematic Behavior of the Native Branch

It is useful to describe how the single native history changes character without interpreting its geometric weights as material components. Define the instantaneous transport fraction of the branch by

$$f_\pi(a) = \frac{\mathcal{T}_\pi a^{-\pi}}{E_\pi(a)^2}. \tag{149}$$

Differentiating Eq. (134) gives

$$\frac{d\ln E_\pi}{d\ln a} = -\frac{\pi}{2} f_\pi(a). \tag{150}$$

A purely kinematic deceleration coordinate can therefore be written

$$q_\pi(a) \equiv -1 - \frac{d\ln H_\pi}{d\ln a} = -1 + \frac{\pi}{2} f_\pi(a). \tag{151}$$

This quantity summarizes the rate evolution of the native branch; it does not introduce a matter or dark-energy decomposition.

In the early transport-dominated limit,

$$q_\pi(a \to 0) \longrightarrow \frac{\pi}{2} - 1 = 0.5707963268, \tag{152}$$

whereas at the present phase

$$q_{\pi,0} = -\frac{1}{2}. \tag{153}$$

The branch crosses $q_\pi = 0$ at

$$a_{\rm tr} = 0.6564607036, \qquad z_{\rm tr} = 0.5233204280. \tag{154}$$

The change from decelerating to accelerating expansion therefore follows from the native π-branch itself. No separate cosmological-constant term is required in the definition of Eq. (134).

4.1.6 Why the Native Branch Must Come First

The branch defined above is the common cosmological source state for every later CPTG comparison layer. It fixes the native phase, rate, acceleration coordinate, and curvature-transport scale before an acoustic, luminosity, CMB, abundance, or distance coordinate is constructed. This is the cosmological analogue of the hierarchy established earlier for

bound systems: first solve or define the native geometric state, then form the reduction required by the physical problem.

The native branch remains a cosmological background construction rather than a generic rescaling law for local bound systems. Redshift selects the native phase and therefore the native quantities associated with that phase; it does not, by itself, evolve or redefine a bound object's source distribution, support state, local structural acceleration, solved field, or post-solution Structural Mode. A change in those local quantities requires either an independently specified physical evolution of the object or an explicitly declared comparison-coordinate operation. The next section therefore turns directly to the downstream observational projections of the native cosmological branch.

5 Observational Projection Hierarchy

5.1 Common-Phase Projection Architecture

The native cosmological branch defined above is the source state of the cosmological construction. Observations, however, do not all interrogate that state through the same physical response channel. Acoustic propagation, luminosity-distance calibration, CMB parameterization, growth, and lensing each expose a different reduction or projection of the same underlying geometry. The purpose of the projection hierarchy is to translate between those channels without allowing the observational coordinates to redefine the native branch [12].

The architecture is therefore

$$\mathfrak{B}_\pi \longrightarrow \{\mathfrak{P}_{\rm ac}, \mathfrak{P}_L\}\,, \tag{155}$$

where $\mathfrak{B}_\pi$ is the one native cosmological history and $\mathfrak{P}_{\rm ac}$ and $\mathfrak{P}_L$ are its acoustic and luminosity projections. They are not independently initialized cosmologies and do not carry independent primordial native normalizations.

The shared geometric phase remains

$$N_\pi = \ln a. \tag{156}$$

For any defined projection b, its rate generates a projection-specific time coordinate through

$$\frac{dN_\pi}{dt_b} = H_b(N_\pi), \qquad H_b\, dt_b = dN_\pi. \tag{157}$$

Thus equal N_π means equal geometric phase, not equal accumulated projection time. The native branch itself generates

$$t_\pi(a) = \int_0^a \frac{da'}{a' H_\pi(a')}, \qquad t_{0,\pi} = 12.7541210323 \text{ Gyr}. \tag{158}$$

The corresponding acoustic and luminosity coordinates are introduced below from their projected rate histories. These integrated projection times are response coordinates generated by the declared rates. The parent theory does not infer from their numerical values that all projected clocks are independent physical proper-time standards.

The distinction between source state and response coordinate is easiest to see by isolating the nonconstant part of the native branch,

$$X_\pi(a) \equiv E_\pi(a)^2 - \mathcal{B}_\pi = \mathcal{T}_\pi a^{-\pi}. \tag{159}$$

There is one such native background function. The first question is therefore not how to fit an acoustic history independently, but whether the acoustic state follows uniquely from X_π. It does.

5.2 Exact Acoustic Projection of the Native Branch

The acoustic sector is the cleanest demonstration of the projection principle because its background mapping is explicit. The acoustic exponent is

$$p_{\rm ac} = 3 - \frac{\pi}{100} = 2.968584073464102, \tag{160}$$

and the corresponding curvature-to-acoustic projection ratio is

$$\mathcal{G}_T = \frac{p_{\rm ac}}{\pi} = 0.9449296585513721. \tag{161}$$

Define the local scalar projector

$$\mathcal{P}_{\rm ac}(X) \equiv \mathcal{T}_\pi^{1-\mathcal{G}_T} X^{\mathcal{G}_T}. \tag{162}$$

Within the declared class of local monomial maps $\mathcal{P}(X) = AX^\gamma$, the two requirements that $\mathcal{T}_\pi a^{-p_C}$ map to exponent $p_{\rm ac}$ and that the coefficient $\mathcal{T}_\pi$ be inherited exactly force $\gamma = p_{\rm ac}/p_C = \mathcal{G}_T$ and $A = \mathcal{T}_\pi^{1-\mathcal{G}_T}$. The projector is therefore uniquely fixed within that declared monomial class; "exact" below refers to the resulting algebraic mapping, not to a claim that no broader non-monomial representation could ever be written.

Applying it to the native nonconstant state gives

$$X_{\rm ac}(a) = \mathcal{P}_{\rm ac}[X_\pi(a)] \tag{163}$$

$$= \mathcal{T}_\pi^{1-\mathcal{G}_T} \left(\mathcal{T}_\pi a^{-\pi}\right)^{\mathcal{G}_T} \tag{164}$$

$$= \mathcal{T}_\pi a^{-p_{\rm ac}}. \tag{165}$$

The acoustic background is therefore

$$E_{\rm ac}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\rm ac}}. \tag{166}$$

The important result is not merely the new exponent. Equation (165) preserves the native coefficient $\mathcal{T}_\pi$ exactly. The acoustic response therefore inherits its primordial background normalization from the native state; there is no additional free acoustic amplitude that must be synchronized with a separately initialized native history. This is the mathematical reason CPTG treats the acoustic sector as a projection rather than a second cosmology [15, 12].

For Hubble-like projected rate coordinates, the fixed CPTG transport rule acts on the present normalization,

$$H_{0,\mathrm{transported}} = H_{0,\mathrm{raw}}\sqrt{\mathcal{G}_T}. \tag{167}$$

Applying this normalization rule to the native present rate gives the associated acoustic coordinate

$$H_{0,\mathrm{ac}}^{\mathrm{CPTG}} = H_0^{(\pi)}\sqrt{\mathcal{G}_T} = 67.4777967351\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{168}$$

This value is fixed by the native normalization and the projection ratio before any CMB comparison is performed. It does not imply that the full acoustic rate history is obtained by multiplying the native history by one constant. Defining

$$H_{\mathrm{ac}}(a) \equiv H_{0,\mathrm{ac}}^{\mathrm{CPTG}} E_{\mathrm{ac}}(a), \tag{169}$$

the relation to the native history is

$$\frac{H_{\mathrm{ac}}(a)}{H_\pi(a)} = \sqrt{\mathcal{G}_T}\,\frac{E_{\mathrm{ac}}(a)}{E_\pi(a)}, \tag{170}$$

which reduces to $\sqrt{\mathcal{G}_T}$ at $a = 1$ because $E_{\mathrm{ac}}(1) = E_\pi(1) = 1$ [12].

The acoustic projected time coordinate is generated by its own rate history,

$$t_{\mathrm{ac}}(a) = \int_0^a \frac{da'}{a' H_{0,\mathrm{ac}}^{\mathrm{CPTG}} E_{\mathrm{ac}}(a')}. \tag{171}$$

For the constant-exponent form above,

$$t_{0,\mathrm{ac}} = 13.88517358\ \mathrm{Gyr}. \tag{172}$$

This is an integrated acoustic response-time coordinate. It is not promoted here to a universal physical age or used to replace the native phase variable.

The acoustic projection establishes how one native background is expressed in the CMB-facing acoustic channel. The luminosity-distance channel asks the complementary question: how does the same native phase appear through the affine luminosity response?

5.3 Affine Luminosity Projection of the Native Branch

The luminosity coordinate is a second projection of the same native π-branch. Unlike the acoustic projection, which changes the exponent of the nonconstant native term, the

registered affine luminosity projection preserves the normalized native shape and changes only the comparison normalization. Its normalization is fixed internally by the CPTG polarization–transport structure rather than by an external luminosity-distance endpoint [12].

The luminosity construction begins with a dimensionless one-ninth response increment. Three algebraically distinct CPTG identities reach the same value. First, the frame-conditional matched isotropic response of the parent $C_{\mathrm{red}}^{1/3}$ polarization sector is

$$\mathsf{C}_{\Pi,\star} = \frac{1}{9} I. \tag{173}$$

Second, the registered structural comparison coefficients satisfy

$$C_R = 48\pi, \qquad C_H = \frac{16\pi}{3}, \qquad \frac{C_H}{C_R} = \frac{1}{9}. \tag{174}$$

These coefficients belong to the structural comparison interface; Eq. (174) does not identify a bound-system curvature scale with the native cosmological curvature scale.

Third, the acoustic comparison weight

$$\Omega_{m,\mathrm{ac}} \equiv \frac{p_{\mathrm{ac}}}{3\pi} \tag{175}$$

and the exact projection ratio $\mathcal{G}_T = p_{\mathrm{ac}}/\pi$ define the acoustic rank quotient

$$\mathcal{R}_3 \equiv \frac{\Omega_{m,\mathrm{ac}}}{\mathcal{G}_T} = \frac{1}{3}, \qquad \mathcal{R}_3^2 = \frac{1}{9}. \tag{176}$$

Here $\Omega_{m,\mathrm{ac}}$ is an acoustic comparison weight, not an additional CPTG matter component. The three theory-side constructions therefore satisfy the common coefficient identity

$$\frac{1}{9} = \frac{C_H}{C_R} = \mathcal{R}_3^2, \tag{177}$$

with the matched polarization response in Eq. (173) carrying the same scalar increment. These are internal algebraic routes to one coefficient; they are not statistically independent measurements.

At the comparison-coordinate level, the registered affine carrier-plus-response map is

$$\mathsf{Q}_L^{\mathrm{aff}} \equiv I + \mathsf{C}_\Pi. \tag{178}$$

At the matched isotropic state,

$$\mathsf{Q}_{L,\star}^{\mathrm{aff}} = \left(1 + \frac{1}{9}\right) I = \frac{10}{9} I, \tag{179}$$

so the affine luminosity normalization is

$$\mathcal{G}_L^{\mathrm{aff}} = \frac{10}{9}. \tag{180}$$

The affine map is a registered comparison-coordinate prescription. It is not a global rescaling of the physical spacetime metric and is not promoted to a universal microscopic transfer operator for arbitrary nonmatched states.

For an isotropic affine comparison co-metric with scalar eigenvalue $q_L = \mathcal{G}_L^{\mathrm{aff}}$, the corresponding effective radius coordinate is

$$R_{C,L}^{\mathrm{eff,aff}}(a) = q_L^{-1/2} R_C(a) = \sqrt{\frac{9}{10}}\, R_C(a). \tag{181}$$

Using the native rate–radius relation, the reciprocal rate coordinate is therefore

$$\frac{H_L^{\mathrm{aff}}(a)}{H_\pi(a)} = \frac{R_C(a)}{R_{C,L}^{\mathrm{eff,aff}}(a)} = \sqrt{\frac{10}{9}}. \tag{182}$$

Define

$$\Gamma_L^{\mathrm{aff}} \equiv \sqrt{\frac{10}{9}}, \qquad \Xi_L^{\mathrm{aff}} \equiv \left(\Gamma_L^{\mathrm{aff}}\right)^{-1} = \sqrt{\frac{9}{10}}. \tag{183}$$

Then the luminosity rate history at common native phase is

$$H_L^{\mathrm{aff}}(a) = \Gamma_L^{\mathrm{aff}} H_\pi(a) = \sqrt{\frac{10}{9}}\, H_\pi(a), \tag{184}$$

with present normalization

$$H_{0,L}^{\mathrm{aff}} = H_0^{(\pi)} \sqrt{\frac{10}{9}} = 73.1711530416 \ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{185}$$

Because the same constant multiplies the complete native rate history and its present normalization, the normalized luminosity history is exactly

$$E_L^{\mathrm{aff}}(a) \equiv \frac{H_L^{\mathrm{aff}}(a)}{H_{0,L}^{\mathrm{aff}}} = \frac{H_\pi(a)}{H_0^{(\pi)}} = E_\pi(a). \tag{186}$$

Thus the acoustic and luminosity projections act differently on one native branch. The acoustic map changes the normalized exponent from π to p_{ac}, whereas the affine luminosity map preserves the native normalized shape exactly and changes only its comparison normalization.

The corresponding affine luminosity comparison clock remains synchronized to the common phase through

$$\frac{dt_L^{\mathrm{aff}}}{dt_\pi} = \Xi_L^{\mathrm{aff}} = \sqrt{\frac{9}{10}}, \qquad H_L^{\mathrm{aff}} dt_L^{\mathrm{aff}} = H_\pi dt_\pi = dN_\pi. \tag{187}$$

With the same zero-time convention as the native branch,

$$t_L^{\mathrm{aff}}(a) = \sqrt{\frac{9}{10}}\, t_\pi(a), \qquad t_{0,L}^{\mathrm{aff}} = 12.0996216046 \ \mathrm{Gyr}. \tag{188}$$

This is a projected comparison-clock coordinate, not a second physical age of the Universe. The native π-branch remains the single cosmological source state throughout.

5.4 Native Horizon Containment and Common Background Origin

Once the native state and exact acoustic projector are separated conceptually, the cosmological horizon question separates into two parts. The spatial question asks whether the physical region associated with the last-scattering sphere fits inside one native curvature-transport domain at the same phase. The projection question asks whether the acoustic background on that domain is uniquely inherited from the native state.

Let $z_\star$ denote the declared last-scattering comparison epoch. The acoustic comparison distance is

$$\chi_\star = \frac{c}{H_0^{(\pi)}} \int_0^{z_\star} \frac{dz}{E_{\rm ac}(z)}, \tag{189}$$

where the dimensional prefactor remains the native normalization while the exact acoustic projection enters through the integrand. This is the registered native-normalized acoustic comparison-distance definition; replacing $H_0^{(\pi)}$ in the prefactor by $H_{0,\rm ac}^{\rm CPTG}$ would define a different comparison coordinate. The physical radius of that future-observable comoving region at a common phase is

$$L_\star(a) = a\chi_\star. \tag{190}$$

The like-for-like native containment coordinate is therefore

$$\Lambda_R(a) \equiv \frac{L_\star(a)}{R_C(a)}. \tag{191}$$

A native background-containment solution at a chosen phase requires $\Lambda_R < 1$; the equality $L_\star = R_C$ defines the corresponding entry phase. Radius is compared with radius throughout, not a full diameter with one native radius.

The background-horizon statement follows only after containment is combined with the exact projector. Within the declared single-native-solution background construction, the connected native domain carries one background value at each common phase. For the local scalar projector of Eq. (162), a homogeneous native background satisfies

$$\nabla_i X_\pi^{(0)} = 0 \quad \Longrightarrow \quad \nabla_i X_{\rm ac}^{(0)} = \mathcal{P}'_{\rm ac}[X_\pi^{(0)}] \nabla_i X_\pi^{(0)} = 0. \tag{192}$$

Thus the acoustic background inherits the native background state algebraically. CPTG does not require separately initialized acoustic patches to exchange heat across a region that is already represented as one connected native background domain. This is a common-origin statement, not a claim of superluminal material propagation, and it is a background-level closure rather than a substitute for a perturbation-level causal-propagation derivation. The numerical containment ratio, entry phase, and associated distances are given in [15].

5.5 CMB-to-BAO Standard-Ruler Coordinates

The exact acoustic projection also fixes how the same projected background is carried into baryon-acoustic-oscillation distance coordinates. The parent theory does not introduce an independently evolving physical rod. The observable object is a dimensionless distance-to-acoustic-scale ratio, while the dimensional acoustic scale used to represent that ratio belongs to the declared comparison coordinate [13].

Define the dimensionless acoustic distance integral

$$I_{\rm ac}(z) \equiv \int_0^z \frac{dz'}{E_{\rm ac}(z')}. \tag{193}$$

At the CMB comparison boundary, let $\theta_\star$ denote the observed angular acoustic scale and $z_\star$ the adopted decoupling endpoint. These are observational comparison inputs rather than native CPTG invariants [4, 13]. The corresponding dimensionless acoustic integral is

$$I_\star \equiv I_{\rm ac}(z_\star). \tag{194}$$

A dimensional acoustic comparison coordinate may then be written as

$$r_{\rm ac} = \theta_\star \frac{c}{H_{0,\rm ac}^{\rm CPTG}} I_\star. \tag{195}$$

The dimensional prefactor is not the quantity carried into the late-time ruler comparison. Dividing the acoustic distance scale by $r_{\rm ac}$ gives the dimensionless normalization

$$A_{\rm CPTG} \equiv \frac{c/H_{0,\rm ac}^{\rm CPTG}}{r_{\rm ac}} = \frac{1}{\theta_\star I_\star}. \tag{196}$$

Thus the common dimensional factor cancels exactly. Once the acoustic shape and the declared CMB boundary are fixed, no late-time survey-specific value of H_0 or dimensional ruler is introduced into the CPTG standard-ruler map.

For a BAO survey whose reported acoustic denominator is denoted generically by r, the CPTG transverse, radial, and isotropic comparison coordinates are

$$\left(\frac{D_M(z)}{r}\right)_{\rm CPTG} = A_{\rm CPTG} I_{\rm ac}(z), \tag{197}$$

$$\left(\frac{D_H(z)}{r}\right)_{\rm CPTG} = \frac{A_{\rm CPTG}}{E_{\rm ac}(z)}, \tag{198}$$

$$\left(\frac{D_V(z)}{r}\right)_{\rm CPTG} = A_{\rm CPTG} \left[\frac{z I_{\rm ac}(z)^2}{E_{\rm ac}(z)}\right]^{1/3}. \tag{199}$$

The anisotropic Alcock–Paczynski shape coordinate removes the common normalization altogether:

$$F_{\rm AP}^{\rm CPTG}(z) \equiv \frac{D_M(z)}{D_H(z)} = E_{\rm ac}(z) I_{\rm ac}(z). \tag{200}$$

These relations are comparison-coordinate consequences of the already fixed native-to-acoustic projection. They do not identify $r_{\rm ac}$ with a separate microscopic CPTG rod, do not add a second cosmological background, and do not permit a BAO dataset to redefine $p_{\rm ac}$, $\mathcal{G}_T$, or the native normalization. The dedicated standard-ruler paper contains the observational covariance tests; the parent theory retains only the coordinate law [13].

5.6 CMB, Growth, and Lensing as Comparison Coordinates

The native background does not by itself supply every conventional variable used by CMB and large-scale-structure likelihoods. CPTG therefore uses a translation layer. In this layer standard scalar-perturbation notation, effective Poisson relations, Weyl-potential combinations, density coordinates, and compressed amplitudes are comparison coordinates for expressing CPTG response in observationally familiar variables. They are not adopted as the governing FLRW equations of the native CPTG branch [12].

For example, using standard scalar-perturbation notation [5], denote the conventional comparison potentials by $\Psi_{\rm cmp}$ and $\Phi_{\rm cmp}$ to avoid collision with the CPTG source potential Φ used earlier. A conventional metric interface is

$$ds^2 = -(1+2\Psi_{\rm cmp})c^2dt^2 + a(t)^2(1-2\Phi_{\rm cmp})d\mathbf{x}^2. \tag{201}$$

The corresponding response functions are defined through

$$k^2\Psi_{\rm cmp} = -4\pi G a^2\bar{\rho}\,\mu_{\rm CPTG}(a,k)\delta, \tag{202}$$

and

$$k^2(\Phi_{\rm cmp}+\Psi_{\rm cmp}) = -8\pi G a^2\bar{\rho}\,\Sigma_{\rm CPTG}(a,k)\delta. \tag{203}$$

These equations define only the comparison interface; they do not replace the parent CPTG field equations. In particular, $\mu_{\rm CPTG}$ and $\Sigma_{\rm CPTG}$ are not free dark-sector functions to be fitted independently against CMB, growth, or lensing data; their role is to express the solved upstream CPTG state in conventional reporting coordinates.

For the registered scale-averaged reduced comparison layer, the branch-derived lensing-response scale is

$$A_L^{\rm CPTG} = \left(\frac{p_C}{p_{\rm ac}}\right)^2 = 1.119956202138. \tag{204}$$

The corresponding fixed compressed growth and lensing responses are

$$\mu_{\rm CPTG}(a) = 1 + \frac{\pi}{2}\left(A_L^{\rm CPTG}-1\right)\left(1-a^{p_{\rm ac}}\right), \tag{205}$$

$$\Sigma_{\rm CPTG}(a) = 1 + \left(A_L^{\rm CPTG}-1\right)\left(1-a^{p_{\rm ac}}\right). \tag{206}$$

These scale-averaged forms organize the registered comparison calculation; they are not presented as a completed native scale-transport derivation and do not convert $\mu_{\rm CPTG}$ or $\Sigma_{\rm CPTG}$ into tunable phenomenological functions [12].

5.6.1 Projection-Law Status

The comparison functions $\mu_{\mathrm{CPTG}}(a,k)$ and $\Sigma_{\mathrm{CPTG}}(a,k)$ are downstream interfaces through which a solved CPTG cosmological response is represented in conventional growth and lensing coordinates. A CMB comparison map is likewise a downstream projection of the fixed native/acoustic construction into the harmonic or likelihood coordinate used by an observational analysis [14]. Every such projection must inherit the same upstream CPTG state and the same upstream geometric law.

Dataset-specific kernels, compressed amplitudes, map constructions, and likelihood settings belong to the child comparison papers. They may implement the projection law, but they do not alter the native branch, the acoustic projector, the upstream CPTG geometry, or the parent response hierarchy, and they provide no posterior retuning channel for those quantities [12, 14].

5.6.2 Authority Boundary of the Comparison Layer

The branch-derived response scale A_L^{CPTG} of Eq. (204) is a CPTG comparison-response coordinate, distinct from the fixed conventional CMB lensing-amplitude setting $A_{\mathrm{lens}} = 1$ used in the comparison likelihood. Bandpower, S_8, map-envelope, survey, and likelihood quantities are downstream reporting coordinates. The parent theory retains only the translation structure required to connect those coordinates to the unchanged native and acoustic branches; their dataset-level evaluation is left to the dedicated comparison papers [12, 14].

The parent claim is narrower and more important conceptually. The hierarchy is

$$\begin{aligned} \text{native CPTG geometry} \quad &\longrightarrow \quad \text{derived projection operators} \\ &\downarrow \quad \text{declared comparison inputs where required} \\ &\longrightarrow \quad \text{derived observational coordinates.} \end{aligned} \tag{207}$$

A conventional variable that appears at the right-hand side of this chain does not thereby become part of the native CPTG matter inventory. A first-principles derivation of perturbative observables directly from the native CPTG geometry would deepen the theory-side derivation of this same upstream state; it would not introduce a tunable dark-matter sector, free growth or lensing functions, or license dataset-by-dataset retuning. The present comparison-coordinate layer is therefore a declared derivational interface, not missing phenomenological freedom. A unique microscopic luminosity-transfer operator remains a separate stronger construction than the affine comparison-coordinate map stated here.

With the cosmological projection hierarchy separated cleanly from the native branch, the same transport–polarization architecture extends into microscopic reaction-state space without replacing the network's native nuclear dynamics. The primordial nuclear sector supplies that realization.

6 Primordial Nuclear-Reaction Geometry

The primordial nuclear realization carries the CPTG polarization–transport architecture into a conserved reaction-state space without replacing microscopic nuclear physics. A native Big Bang nucleosynthesis (BBN) network retains its own rates, weak processes, thermodynamics, Jacobians, and stiff dynamical evolution. CPTG supplies the upstream scalar response and exact reaction-source geometry; the native network supplies the dynamical pushforward from that source to the abundance observables [16].

The parent hierarchy is therefore

$$\begin{aligned} &\text{fixed CPTG scalar response} \longrightarrow \text{exact reaction-source geometry} \\ &\qquad\qquad\qquad\qquad\quad \downarrow \ \text{network-native dynamical pushforward} \\ &\qquad\qquad\qquad\qquad\quad \longrightarrow \text{abundance-space projection.} \end{aligned} \tag{208}$$

The first two layers define the CPTG nuclear law. The third remains native to the selected reaction network, and the final abundance coordinate is a downstream readout. This ordering forbids an endpoint abundance from redefining the source geometry or scalar law that generated it.

6.1 Spacetime-to-Nuclear Bridge

The nuclear reaction-source basis is not identified with four literal tensor components of $\Pi^{\mathrm{pol}}_{\mu\nu}$ or $\Theta^{\mathrm{trans}}_{\mu\nu}$. The shared CPTG content is the scalar geometric response inherited from the native and acoustic cosmological construction, while nuclear stoichiometry supplies the exact baryon- and charge-conserving source directions that are specific to the reaction state space. Schematically,

$$\underbrace{(\mathcal{G}_T, J_{\mathrm{CPTG}}, K_{\mathrm{CPTG}})}_{\text{shared CPTG scalar geometry}} \quad + \quad \underbrace{\mathbf{s}_r}_{\text{stoichiometric source}} \quad \xrightarrow{\mathcal{N}} \quad \underbrace{\mathbf{y}_{r,\mathcal{N}}}_{\text{network-native response}} . \tag{209}$$

Thus the CPTG-specific statement is not the rank of stoichiometry by itself. It is the coupling of the conserved source directions to the fixed scalar geometry before the native dynamical pushforward. The reaction network remains a physical nuclear system in its own right; CPTG does not replace its microscopic rates or solver dynamics.

6.2 Primordial Coordinate and Scalar Curvature Law

The nuclear realization inherits the acoustic transport ratio

$$\mathcal{G}_T = \frac{p_{\mathrm{ac}}}{p_C} = \frac{3 - \pi/100}{\pi} = 0.9449296585513721. \tag{210}$$

The fixed acoustic baryon-to-photon reference coordinate is

$$\eta_{10}^{\mathrm{ac}} = 6.17038016771, \tag{211}$$

and the transported primordial comparison coordinate is

$$\eta_{10}^{\mathrm{BBN}} = \eta_{10}^{\mathrm{ac}}\sqrt{\mathcal{G}_T} = 5.998071834744. \tag{212}$$

The displayed values of η_{10}^{ac} and $\mathcal{G}_T$ are rounded publication values, whereas the fixed η_{10}^{BBN} coordinate is evaluated from the full stored precision of the upstream transport quantities. Recomputing from the displayed values gives 5.998071834741609; the 2.39×10^{-12} difference from Eq. (212) is therefore display-rounding provenance, not a distinct physical correction or fitted adjustment [18].

These two coordinates have different roles. The superscripted quantities in Eqs. (211) and (212) are fixed comparison coordinates. An unadorned η_{10} below denotes the local baryon-density coordinate at which the reaction operator is evaluated; it is not automatically identified with η_{10}^{BBN}.

For a multiplicative perturbation f_r of reaction r, define the symmetric logarithmic coordinate

$$x = -\ln f_r. \tag{213}$$

The fixed first- and second-order scalar coefficients are

$$J_{\mathrm{CPTG}}(\eta_{10}) = \frac{1}{\sqrt{8}}\sqrt{\frac{\eta_{10}}{\eta_{10}^{\mathrm{ac}}}}, \tag{214}$$

and

$$K_{\mathrm{CPTG}}(\eta_{10}) = -\frac{3}{16}\frac{\eta_{10}}{\eta_{10}^{\mathrm{ac}}}\mathcal{G}_T^{-1/3} = -\frac{3}{2}J_{\mathrm{CPTG}}^2\mathcal{G}_T^{-1/3}. \tag{215}$$

Writing

$$s_T \equiv \mathcal{G}_T^{1/3}, \tag{216}$$

the compact scalar generator is

$$g_T(x;\eta_{10}) = \frac{s_T}{3}\ln\left(1 + \frac{3J_{\mathrm{CPTG}}(\eta_{10})}{s_T}x\right). \tag{217}$$

Its local expansion begins

$$g_T(x) = J_{\mathrm{CPTG}}x + K_{\mathrm{CPTG}}x^2 + \mathcal{O}(x^3), \tag{218}$$

with admissible perturbative domain

$$1 + \frac{3J_{\mathrm{CPTG}}(\eta_{10})}{\mathcal{G}_T^{1/3}}x > 0. \tag{219}$$

The first- and second-order coefficients in Eqs. (214) and (215) are the registered scalar law. Higher coefficients generated by Eq. (217) are analytic continuations of that generator and are not promoted here to independent universal constants.

6.3 Four-Sector Baryon State

The light-nuclear backbone is organized as

$$(n, p) \to \mathrm{D} \to (T, {}^3\mathrm{He}) \to {}^4\mathrm{He}, \tag{220}$$

with structural roles

$$\text{vertex} \to \text{bridge} \to \text{closure} \to \text{saturation}. \tag{221}$$

The vertex is the free neutron–proton source population, deuterium is the first bound bridge, the tritium–helium-three mirror pair is the charge-resolved mass-three closure sector, and helium four is the saturation sector of this declared light-element state space. Saturation does not mean that nuclear evolution above helium four is forbidden; it identifies the terminal structural sector of the exact four-sector backbone.

The baryon-weighted state is

$$\mathbf{X} = \begin{pmatrix} X_V \\ X_B \\ X_C \\ X_S \end{pmatrix} = \begin{pmatrix} Y_n + Y_p^{\mathrm{free}} \\ 2Y_D \\ 3(Y_T + Y_{^3\mathrm{He}}) \\ 4Y_{^4\mathrm{He}} \end{pmatrix}. \tag{222}$$

The internal charge-polarization coordinates are

$$P_V = Y_p^{\mathrm{free}} - Y_n, \qquad P_C = 3(Y_{^3\mathrm{He}} - Y_T). \tag{223}$$

The six coordinates $(X_V, P_V, X_B, X_C, P_C, X_S)$ reconstruct the six declared species exactly; the corresponding linear coordinate transformation has determinant 288. The four-sector representation therefore becomes lossless for this six-species state once the two internal polarization coordinates are retained.

6.4 Ordered Transport and Charge-Constrained Polarization

Transport between adjacent structural sectors is generated by

$$\mathsf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \frac{d\mathbf{X}}{dx} = \mathsf{D}\mathbf{j}, \tag{224}$$

with

$$\mathbf{j} = (j_{VB}, j_{BC}, j_{CS})^T. \tag{225}$$

Because

$$\mathbf{1}^T \mathsf{D} = 0, \tag{226}$$

the ordered transport operator preserves total baryon number exactly. The three independent columns of D span the vertex-to-bridge, bridge-to-closure, and closure-to-saturation transport directions.

Charge conservation supplies the independent internal polarization direction. Every baryon- and charge-conserving perturbation in the declared six-species sector satisfies

$$3\,\delta P_V + \delta P_C = 0, \tag{227}$$

so one independent polarization coordinate remains:

$$P_{\mathrm{nuc}} = \delta P_V = -\frac{\delta P_C}{3}. \tag{228}$$

The universal reaction-source coordinate is therefore

$$\mathbf{s}_r = \begin{pmatrix} q_{VB} \\ q_{BC} \\ q_{CS} \\ P_{\mathrm{nuc}} \end{pmatrix}_r. \tag{229}$$

Since $\operatorname{rank}\mathsf{D} = 3$ and charge conservation leaves one independent polarization mode,

$$\mathcal{S}_{\mathrm{nuc}} = \mathcal{T}_{\mathrm{nuc}} \oplus \mathcal{P}_{\mathrm{nuc}}, \qquad \dim \mathcal{S}_{\mathrm{nuc}} = 3 + 1 = 4. \tag{230}$$

This rank-four source space is exact within the declared free-nucleon, deuterium, mass-three, and helium-four state. Reaction stoichiometry selects $\mathbf{s}_r$; it does not introduce reaction-specific geometric coefficients.

6.5 Network-Native Dynamic Observable Pushforward

The universal source geometry does not imply a universal static endpoint matrix shared by all nuclear codes. Each admissible network $\mathcal{N}$ retains its own native rates, thermodynamics, currents, Jacobians, integration history, and numerical realization. The CPTG law enters that network through the source current and the fixed source coordinate.

Let $\mathcal{N}_{r,\mathcal{N}}(x)$ denote the baseline-subtracted integrated net event current of reaction r in network $\mathcal{N}$. Locally,

$$\mathcal{N}_{r,\mathcal{N}}(x) = \mathcal{A}^{(1)}_{r,\mathcal{N}} x + \mathcal{B}^{(2)}_{r,\mathcal{N}} x^2 + \mathcal{O}(x^3), \tag{231}$$

where

$$\mathcal{A}^{(1)}_{r,\mathcal{N}} = \left.\frac{d\mathcal{N}_{r,\mathcal{N}}}{dx}\right|_{x=0}, \qquad \mathcal{B}^{(2)}_{r,\mathcal{N}} = \frac{1}{2}\left.\frac{d^2\mathcal{N}_{r,\mathcal{N}}}{dx^2}\right|_{x=0}. \tag{232}$$

The direct first-order source drive is

$$\mathbf{d}^{(1)}_{r,\mathcal{N}} = \mathcal{A}^{(1)}_{r,\mathcal{N}} \mathbf{s}_r. \tag{233}$$

A local first-order representation of the native dynamical pushforward may be written

$$\mathbf{u}^{(1)}_{r,\mathcal{N}} = \boldsymbol{\chi}^{(1)}_{\mathcal{N}}(\eta_{10}) \mathbf{d}^{(1)}_{r,\mathcal{N}}, \tag{234}$$

where $\mathbf{u}^{(1)}_{r,\mathcal{N}}$ is the four-coordinate observable response and $\boldsymbol{\chi}^{(1)}_{\mathcal{N}}$ is a local representation of the network-native dynamic response. It is not a universal static endpoint matrix and is not required to be numerically identical across independently implemented networks.

For the registered single-reaction second-order local representation,

$$\mathbf{u}^{(2)}_{r,\mathcal{N}} = \boldsymbol{\chi}^{(1)}_{\mathcal{N}}(\eta_{10}) \left(\mathcal{B}^{(2)}_{r,\mathcal{N}} \mathbf{s}_r\right) + \frac{K_{\mathrm{CPTG}}(\eta_{10})}{J_{\mathrm{CPTG}}(\eta_{10})} \mathbf{u}^{(1)}_{r,\mathcal{N}}. \tag{235}$$

The first term carries the native source-current curvature through the local network response, while the second carries the universal CPTG scalar curvature into the already propagated first-order response. Mixed simultaneous multi-reaction derivatives are not part of this single-channel operator. Detailed code-specific response kernels and numerical realizations belong to the dedicated nuclear papers [16].

6.6 Mass-Seven Live Abundance Transport

Mass seven remains separate from the exact four-sector simplex. The light-sector theorem is not enlarged by adding a fifth structural coordinate for lithium. Instead, the CPTG mass-seven law acts directly on the live ^{7}Li and ^{7}Be species [17, 18].

Define

$$Y_7 = Y_{^7\mathrm{Li}} + Y_{^7\mathrm{Be}}. \tag{236}$$

The live source-network transport term is

$$\left.\frac{dY_{^7\mathrm{Li}}}{dt}\right|_{\mathrm{CPTG}} = \left.\frac{dY_{^7\mathrm{Li}}}{dt}\right|_{\mathrm{native}} - \Gamma_7(t) Y_{^7\mathrm{Li}}, \tag{237}$$

$$\left.\frac{dY_{^7\mathrm{Be}}}{dt}\right|_{\mathrm{CPTG}} = \left.\frac{dY_{^7\mathrm{Be}}}{dt}\right|_{\mathrm{native}} - \Gamma_7(t) Y_{^7\mathrm{Be}}. \tag{238}$$

Its fixed integrated optical depth is

$$\tau_7 = \int \Gamma_7(t)\, dt = \ln \pi, \qquad e^{-\tau_7} = \frac{1}{\pi}. \tag{239}$$

The factor $1/\pi$ is the exact isolated-channel survival implied by the integrated transport law. In a live reaction network, simultaneous native production and destruction remain active, so the endpoint abundance is determined by the full network evolution rather than by a post-processing division by π.

The radiation-side primordial comparison coordinate is

$$N_{\mathrm{eff}}^{\mathrm{CPTG}} = p_{\mathrm{ac}} = 3 - \frac{\pi}{100} = 2.968584073464. \tag{240}$$

It is an effective relativistic-degree comparison coordinate, not an additional native radiation species. The primordial comparison tuple is therefore

$$\Theta_{\mathrm{BBN}}^{\mathrm{CPTG}} = \left(\eta_{10}^{\mathrm{BBN}}, N_{\mathrm{eff}}^{\mathrm{CPTG}}, \tau_7\right), \tag{241}$$

with each entry retaining the authority class fixed by its defining construction.

6.7 Heavy-Sector Continuation Boundary

The exact universal theorem in this parent paper is the four-sector light-nuclide source geometry above. Extension into heavier mass sectors is a downstream computational continuation, not an additional axiom of the rank-four theorem. In the associated PRIMAT-anchored construction, the native primordial inventory supplies trajectory and endpoint information through $A = 23$, after which a declared seed-free prescribed-bath reduced transport graph defines a continuous mass-sector register through

$$A = 119. \tag{242}$$

The external Y_A values of that continuation are prescribed-bath reduced-transport populations. They are not self-consistent final abundances of a coupled heavy-element BBN network and are not asserted primordial heavy-element yield predictions. The reduced-graph calculation therefore demonstrates how the fixed CPTG source architecture can be continued computationally beyond the light theorem while preserving a strict boundary between structural reachability and abundance physics [19].

The nuclear sector consequently realizes the parent CPTG hierarchy as a fixed scalar law, an exact rank-four conserved source geometry, a network-native dynamical pushforward,

and a separate live mass-seven transport law. Numerical network comparisons and implementation evidence remain in the child nuclear papers rather than being duplicated in this parent theory.

7 Cross-Regime Consistency, Authority, and Falsifiability

CPTG is a framework of scale-dependent realizations with one parent authority discipline. The reduced equations need not share identical coordinates, but they must preserve the source–response ordering, conservation requirements, and common-limit behavior fixed by the parent theory.

7.1 Cross-Regime Consistency Laws

The parent requirements are:

1. **Source and conservation consistency.** Ordinary physical stress–energy remains the gravitational material source, the geometric response satisfies Eq. (12), and non-gravitational state-space realizations preserve their own native conservation laws.

2. **Declared realization.** The source, scale, symmetry, dynamical state, and boundary or state data are specified before the operative reduced law is solved.

3. **Overlap consistency.** Shared physical quantities agree wherever two realizations possess a genuine common physical domain, according to Eq. (2).

4. **GR recovery.** Wherever curvature polarization and curvature transport become negligible, the gravitational solution approaches the Einstein limit of Eq. (9) for the same physical source.

5. **Structural authority.** Local support coordinates such as $a_\star$, regime-specific derived scales, and post-solution fingerprints such as Structural Mode N retain only the authority assigned by their defining realization. No galaxy, cluster, or compact structural coordinate is promoted automatically to a universal cross-regime parameter.

6. **Projection inheritance.** Acoustic, luminosity, CMB, BAO, growth, and lensing coordinates inherit one native cosmological state and remain downstream of that state.

7. **Native dynamical pushforward.** In the nuclear realization, CPTG fixes the geometric source construction while the selected native reaction system performs the time evolution and preserves the applicable baryon and charge constraints.

8. **Finite compact admissibility.** A completed compact state is finite in its physical interior, preserves the declared exterior gravitational charges, and joins the GR-controlled exterior continuously in the common regime.

9. **No cross-regime coefficient borrowing.** A coefficient, continuation rule, or observational projector registered in one realization is not transferred into another merely because the same parent polarization–transport language appears in both.

Failure of one of these conditions is a failure of the corresponding realization to remain consistent with the parent CPTG hierarchy.

7.2 Scope and Derivational Boundaries

The compact equations stated here close the spherical finite mass state and a conditional mass–spin handoff; Kerr-compatible rotating Structural Mode, trapping, clock, and frame-dragging quantities are downstream readouts of that closed state rather than a separately derived general axisymmetric interior field. The anisotropic merger realization is represented by the declared directional transport operator; a general fully covariant time-dependent closure is not supplied by that reduced operator alone. The affine luminosity law is a registered comparison-coordinate map of the native cosmology and is not identified with a unique microscopic luminosity-transfer mechanism. In the nuclear sector, the exact four-sector source geometry and native-network pushforward are distinct from the prescribed-bath reduced continuation above the native primordial network boundary; the latter establishes structural transport under its declared operator rather than self-consistent heavy-element primordial abundances.

These statements define the authority of the equations in this article. A stronger claim requires an additional derivation appropriate to that stronger physical problem; it is not obtained by extrapolating a reduced law outside its declared domain.

8 Conclusion

Curvature Polarization Transport Gravity is organized by one parent rule: physical sources generate geometry, curvature polarization reorganizes that sourced field, curvature transport carries or retains the organized response, and each downstream readout is constructed only after the appropriate physical realization has been declared and solved, while external comparison inputs retain their explicitly declared input status. General Relativity remains the common gravitational limit wherever the additional CPTG response becomes negligible.

For resolved bound systems, the operative law is local to the measured support state. Disk galaxies use the direct baryonic source and radius-dependent support coordinate; pressure-supported clusters use the spherical support realization of the same resolved source–polarization–transport architecture. The separately registered analytic weak-field galaxy reduction yields the fourth-power baryonic Tully–Fisher scaling in its declared intermediate regime and the fixed-outer-source asymptote $V \propto r^{1/4}$ in the far-outer continuation. These analytic limits are not substituted for the measured-domain resolved law.

Structural Mode provides a common post-solution language for gravitational organization. Its parent definition is $N = R/\lambda$, while λ is constructed from the solved geometry appropriate to the object. Galaxies, spherical clusters, and finite rotating compact states therefore carry distinct organization operators but the same physical status: N is an object-specific fingerprint measured after gravity has been solved. CSMI remains a downstream galaxy catalog representation of that continuous fingerprint.

Anisotropic systems retain directional transport rather than being forced into a spherical or disk reduction. Compact gravity closes a finite spherical mass state, separates localized physical source mass from the geometric share of the conserved exterior gravitational charge through the conditional mass–spin handoff, preserves ordinary horizons, and joins the Kerr or Schwarzschild exterior in the GR-controlled regime. Rotating structural quantities are evaluated downstream of that compact closure.

Cosmology contains one native π-branch. The exact acoustic projection, affine luminosity map, CMB comparison coordinates, CMB-boundary standard-ruler construction, BAO coordinates, growth variables, and lensing variables all inherit that native state. Their purpose is to translate one CPTG cosmology into the coordinates used by observation; they do not introduce a second expansion history or an additional cosmological component.

The primordial nuclear realization carries the same polarization–transport organization into a conserved reaction-state space rather than applying the gravitational field equation to nuclear dynamics. The light-nuclide backbone is the ordered vertex–bridge–closure–saturation structure, exact reaction stoichiometry supplies three transport directions plus one charge-constrained polarization direction, and the native reaction network performs the dynamical pushforward. The separate mass-seven law acts as a live source-network transport term, while the heavier reduced continuation retains its explicitly narrower authority.

CPTG is therefore defined less by one formula written at every scale than by a common law of physical authority: declare the source and geometry, use the registered realization for that regime, preserve its conservation laws, require agreement in genuine overlap limits, recover GR where the additional geometric response vanishes, solve the physical state, and only then construct downstream structural or observational coordinates.

Data and Code Availability

This parent article states the governing CPTG laws and authority boundaries. Detailed observational tests, network campaigns, numerical diagnostics, and reproducibility evidence are retained in the cited companion publications and their evidence records. Supporting source code, companion manuscripts, and public research resources are maintained in the CPTG repository [20].

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